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Bachelor Thesis

Exploring the Viability of Google Takeout Data for Mental Health Monitoring

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ABSTRACT

Suicide risk prediction frequently fails in many cases because clinical monitoring tools rely on the active cooperation of the patient. This creates in many cases unreliable data and important missing values. This thesis evaluates whether Google Takeout, the passive data archive available for any Google account, can be used as a reliable passive digital phenotyping source for detecting pre-suicide behavioral signals.

The study applies a three-step Bayesian pipeline (Local Observation Model → Heterogeneous Mixture Model → Bayesian Online Change-Point Detection) to data from 31 patients with a confirmed event of suicide risk. Prospective performance was estimated via nested Leave-One-Patient-Out Cross-Validation (LOPOCV).

A signal is present. Best in-sample discrimination reaches AUC 0.897, and LOPOCV sensitivity at the clinically critical threshold of ≤ 2 alarms/month achieves **20.0%** (6/30 patients). The system performance flattens at 36.7% at 5 alarms/month, which suggests that around a third of the cohort can be truly 'detectable' at any reasonable false-alarm level. In some stages, Takeout data has been benchmarked against EB2, a professional digital phenotyping platform, and preliminary results show that Takeout can achieve similar results even when lacking important variables such as sleep time.

Takeout-based monitoring cannot replace clinical follow-up (roughly four out of five attempts will still be missed at the operating level of alarms per month), but it requires no patient effort and can serve as a zero-cost triage layer identifying the $\sim 16\%$ of patients whose digital data can be used as a predictive signal. The crucial limitation in this thesis is cohort size ($n = 31$). This thesis is embedded in a larger project that is still being developed and whose cohort is being expanded, so all findings will need to be confirmed as the cohort grows.

Keywords: Bayesian Online Change-Point Detection (BOCPD); Heterogeneous Mixture Models (HetMM); Local Observation Model (LOM); Google Takeout; suicide prevention; digital phenotyping; passive sensing.

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LIST OF ABBREVIATIONS

AEPD	Agencia Española de Protección de Datos
AUC	Area Under the (ROC) Curve
BOCPD	Bayesian Online Change-Point Detection
CI	Confidence Interval
CPD	Change-Point Detection
EB2	Evidence-Based Behaviour. Research-grade active digital phenotyping platform used as the clinical benchmark in this study.
EMA	Ecological Momentary Assessment
FN	False Negative
FPR	False Positive Rate
FROC	Free-Response Receiver Operating Characteristic
GDPR	General Data Protection Regulation
GPS	Global Positioning System
HetMM	Heterogeneous Mixture Model
LOM	Local Observation Model
LOPDGDD	Ley Orgánica de Protección de Datos Personales y Garantía de los Derechos Digitales
LOPOCV	Leave-One-Patient-Out Cross-Validation
MDR	Medical Device Regulation (EU 2017/745)
MNAR	Missing Not At Random
ROC	Receiver Operating Characteristic
SaMD	Software as a Medical Device
SNR	Signal-to-Noise Ratio
TFG	Trabajo Fin de Grado (Bachelor's Thesis)
TP	True Positive

Variable codes used throughout the report:

S	Steps (physical activity intensity)
A	App Usage (binary screen-engagement flag)
D	Distance (GPS displacement magnitude)
G	Geolocation (mobility frequency)
K	Weekend flag (temporal confounder control)
V	Physical Activity (EB2 only)
Z	Sleep (EB2 only)
E	Emotions (EB2 only)
R	Sport (EB2 only)
W	Withdrawal flag (derived: density drop below personal baseline)

1. INTRODUCTION

1.1. The problem

Suicide is both one of the most preventable causes of death and one of the most difficult to predict. Meta-analyses of fifty years of risk-assessment research show that simple questionnaires and standardized risk scales perform only marginally better than chance at identifying patients who will attempt suicide [1]. This makes complete sense when one considers the underlying reality: people who are seriously considering an attempt usually hide their intentions from family members and clinicians. The most reliable risk factors to date are only stable on a scale of years (prior attempts, psychiatric diagnoses, demographic features) and cannot provide acute alerts in the days preceding an attempt.

This creates a mismatch between *when* clinical assessment happens (at appointments that may be weeks or months apart) and *when* risk peaks (in the hours and days preceding an attempt). Closing this gap requires continuous monitoring. However, any clinical sensing platform whose efficacy depends on continued patient cooperation is structurally vulnerable to a missing-not-at-random (MNAR) failure mode, in which the signal disappears at the exact moment it is most needed.

1.2. Digital phenotyping and the passive monitoring opportunity

The field of *digital phenotyping* has emerged as a response to this monitoring gap. Digital phenotyping describes the real-time, high-frequency characterization of individual behavior using data generated by personal digital devices [2]. The base idea is that smartphones produce a continuous trace of everyday behavior (location histories, application usage patterns, step counts) that may encode mental state signals at a temporal resolution that no scheduled clinical encounter can match.

Two broad implementations exist. *Active* digital phenotyping platforms prompt patients to log mood, complete brief questionnaires, and grant access to sensor streams. These platforms have demonstrated the feasibility of behavioral monitoring for mental health, but they suffer from the same structural vulnerability described above: patients who stop engaging are precisely those most at risk [3]. *Passive* digital phenotyping collects data automatically as a byproduct of normal smartphone use, without any patient action, and is therefore immune to this MNAR failure mode.

Google Takeout offers a specific instantiation of passive digital phenotyping. It is the consumer data archive that any Google account holder can request, containing location history (from Google Maps and Location Services), step counts (from Google Fit and connected fitness devices), and application usage timestamps (from Android system logs).

Three properties make it especially attractive for suicide risk monitoring:

1. **Zero ongoing effort.** Once a patient grants export consent, the data accumulates indefinitely and automatically, without requiring active patient engagement.
2. **High temporal resolution.** GPS data and application-session logs are recorded at minute-to-hour granularity, which is denser than any clinical instrument can achieve.
3. **Population scale.** Roughly two-thirds of adults carry an Android device, and the data is already being collected by Google regardless of any research study. The marginal cost of using it is zero.

The cost of these advantages is a narrower set of variables than a dedicated clinical platform can capture, specifically, Takeout provides four behavioral dimensions (Steps, App Usage, Distance, and Geolocation) but not the sleep quality, mood ratings, or exercise logs that clinical platforms collect. To bound what passive monitoring can achieve, this study benchmarks Takeout against **EB2**, a research-grade digital phenotyping platform used by the clinical group that provided the data. The central research question is not just whether Takeout contains any behavioral signal, but whether it constitutes a *viable* passive data source for suicide risk monitoring, and how it compares to an already-proved clinical digital phenotyping database like EB2.

It should also be noted that the use of passively collected personal data for health monitoring carries ethical, legal, and regulatory obligations, including compliance with the General Data Protection Regulation (GDPR). These aspects are addressed in Chapter 9 and Chapter 10.

1.3. The core hypothesis

This project rests on the following hypothesis:

Suicide attempts are preceded by statistically detectable deviations from each patient's habitual daily behavioural routine, and these deviations are present in passively collected smartphone data at a temporal resolution of days.

This hypothesis decomposes into three independent claims:

1. **A behavioural signature exists.** Patients' daily routines change in the days preceding an attempt.
2. **The signature is captured by Takeout's variables.** The four passive variables Takeout provides carry enough of the signal to be useful.

3. **The signature is detectable at a clinically tolerable false-alarm rate.** A monitoring system that generates an alarm every two or three days would be clinically useless regardless of its sensitivity.

1.4. Research objectives

The specific objectives of this study are:

1. **Estimate prospective sensitivity** at multiple operating points along the false-alarm-rate axis (1.5, 2, 3, 4, 5, and 6 alarms per month per patient), using nested Leave-One-Patient-Out Cross-Validation (LOPOCV).
2. **Characterise the variable importance hierarchy**, including pairwise synergies and redundancies.
3. **Quantify uncertainty** at the main conclusions. Given the current small cohort, key results are reported using confidence intervals, making a clear distinction between findings that are confirmed and those that are promising but lack sufficient evidence to rule out the null hypothesis.
4. **Assess the overall viability** of Google Takeout as a passive monitoring tool for suicide risk, and determine the conditions under which it could serve as a complementary triage layer in clinical practice.

1.5. Report structure

The remainder of this thesis is organised as follows. Chapter 2 provides the theoretical background, beginning with the concept of digital phenotyping and the active/passive data source distinction (EB2 vs Google Takeout), then describing the BehaviorCPD pipeline (LOM $\rightarrow$ HetMM $\rightarrow$ UnionCPD) and its algorithmic core (Bayesian Online Change-Point Detection), and closing with the evaluation metrics framework and a review of the relevant literature. Chapter 4 describes the cohort, the data acquisition and preprocessing pipeline, the full experimental design, and the methodological corrections applied in the analysis. Chapter 5 presents the results. Chapter 6 interprets those findings against each research objective and addresses the central deployment-viability question. Chapter 7 states the conclusions with their associated evidence levels. Chapter 8 discusses sample-size limitations, methodological caveats, and the next steps. The regulatory and socio-economic implications of deploying a Takeout-based monitoring system are addressed in Chapter 9 and Chapter 10.

2. BACKGROUND AND THEORETICAL FRAMEWORK

This chapter provides the conceptual scaffolding required to understand the methodology and results of the thesis. It opens with the concept of digital phenotyping and the distinction between active and passive data sources, then introduces the two data sources used in this study (EB2 and Google Takeout), describes the BehaviorCPD pipeline, presents the algorithmic core (Bayesian Online Change-Point Detection), and closes with the evaluation metrics framework.

2.1. Digital phenotyping

The term *digital phenotyping* was introduced by Insel [2] to describe the real-time, high-frequency characterization of individual behavior using data generated by personal digital devices. The core premise is that smartphones produce continuous data that may encode the mental situation of the user. Early studies demonstrated that passive smartphone features such as GPS trajectories, phone call logs, and screen-on time can correlate with self-reported depression severity in college students [4], and that location and activity data can discriminate between mood states in bipolar disorder [5].

Active digital phenotyping platforms ask patients to contribute data explicitly: respond to surveys, log mood, and grant access to sensor streams. Key platforms in this space include AWARE [6], an open-source mobile sensing framework widely used in clinical mental health research, and CrossCheck [7], which combined passive sensing with ecological momentary assessment (EMA) in schizophrenia populations and found that passive data quality was consistently superior to EMA data quality during symptom-exacerbation periods. For suicide specifically, Kleiman et al. [8] showed that short-term fluctuations in suicidal ideation follow identifiable within-person patterns that are not captured by weekly or monthly assessments, suggesting the relevant temporal scale for monitoring is hours-to-days rather than weeks-to-months. However, a systematic review by Torous et al. [3] found that active-sensing app engagement rates drop below 50% within the first month in most clinical populations, and that patients who disengage earliest are typically the most symptomatic.

Passive digital phenotyping addresses this limitation by collecting data without requiring any patient action. Studies using passive location logs have shown that GPS mobility features predict self-reported depressive symptoms better than daily questionnaires [9], and that GPS-derived mobility regularity correlates with depression scores [10]. Passive approaches are immune to deliberate non-response, but they rely on consumer-grade data archives whose variable coverage and signal quality were not originally designed for clinical use. Whether such archives contain a usable clinical signal is the central question of this thesis.

2.2. Data sources: EB2 and Google Takeout

2.2.1. EB2: a research-grade platform

EB2 is the research-grade digital phenotyping platform used by the clinical group that provided the data for this study. It collects data through a dedicated application installed on participants' smartphones, combining passive sensor streams (step counts, location, application usage, sleep) with active patient-reported variables (mood ratings, exercise logs). EB2 contributes nine variables to the pipeline (S, A, D, G, V, Z, E, R, K). In this study, EB2 serves as the reference benchmark: Takeout-based results are evaluated against what a dedicated active platform achieves on the same cohort.

2.2.2. Google Takeout: a passive consumer archive

Google Takeout is the data export service that any Google account holder can request. It contains location history (from Google Maps and Location Services), step counts and physical activity (from Google Fit and connected fitness devices), and application usage timestamps (from Android system logs). The data accumulates as a byproduct of normal smartphone use with no ongoing patient effort.

The cost of this advantage is a narrower variable palette: Takeout provides reliable signals for four behavioral dimensions: physical activity (Steps, S), digital engagement (App Usage, A), mobility magnitude (Distance, D), and mobility frequency (Geolocation, G). However, important variables are missing, such as the sleep, mood, or exercise data that EB2 collects.

2.3. Variable codebook

The pipeline manages nine different variables from both Takeout and EB2. Each variable has been assigned a single letter for simplicity. This 'dictionary' is the one that has also been followed in the code of the project. The codebook below shows the letter convention used throughout this report, along with a short description of what each variable measures and how it has been processed.

It is important to highlight that all variables are processed in 30-minutes summary bins, resulting in 48 data points for a day.

2.3.1. Variables shared by Takeout and EB2

ID	Name	Processing	Behavioral meaning
S	Steps	$\log(\text{sum}+1)$ per 30-min bin	Measures physical activity (the number of steps). The log transform handles the heavy right tail of step distributions.
A	App Usage	Binary 0/1 per bin	Whether the patient interacted with their phone during the bin (screen engagement).
D	Distance	$\log(\text{sum}+1)$ per bin	Distance traveled (meters) without differentiating if it was by vehicle or walking.
G	Geolocation	Binary 0/1 per bin	Whether the location sensor triggered a sample during the bin. This is not location coordinates but the temporal pattern of when the sensor was active. Google Location History stores new coordinates when the user moves from one point to another.

2.3.2. EB2-only variables

ID	Name	Processing	Behavioral meaning
V	Activity	$\log(\text{sum}+1)$	Walk/run/vehicle detection from the EB2 activity classifier.
Z	Sleep	Normalized 0–1	Sleep duration and start time, normalized as a fraction of a 24-hour day.
E	Emotions	Raw scalars	Patient-reported mood ratings.
R	Sport	Binary 0/1	Exercise sessions logged by the patient.

2.3.3. Generated variables

ID	Name	Processing	Behavioral meaning
K	Weekend flag	Binary 0/1	1 on weekends, 0 on weekdays. K is not a behavioral signal.
W	Withdrawal flag	Binary 0/1	A data-quality indicator that measures App Usage withdrawal. It flags if the fraction of active app usage in the last 3 days drops below the 30-day baseline by at least 30%.

2.3.4. The Distance-Geolocation distinction

The split between D (Distance) and G (Geolocation) deserves additional explanation because it captures a behavioral insight that became clear only late in the analysis. D measures *how far* the patient is moving in meters. G measures *how often* the location sensor

records a sample, the temporal frequency of mobility events. These are highly correlated for active patients but decouple in informative ways:

- A patient who drives 50 km to a single destination, parks, and stays there has high D (50,000) but low G after arrival.
- A patient who walks around their neighborhood throughout the day has moderate D and high G (continuously changing position).
- A patient who stays home all day has low D and low G.

2.4. The BehaviorCPD pipeline

Raw smartphone data is not fed directly into a change-point detector. The pipeline includes two intermediate stages that compress each patient-day into a low-dimensional representation before the change-point alarm is emitted. The three stages are: Local Observation Model (LOM) → Heterogeneous Mixture Model (HetMM) → UnionCPD aggregator.

2.4.1. Stage 1: Local Observation Model (LOM)

The LOM bins each variable’s raw event stream into 30-minute time slots across the 24-hour day, producing for each patient-day a matrix of shape (variables $\times$ 48 bins). This aggregation converts irregular event streams (GPS pings at unpredictable times, step-count accumulations at irregular intervals) into a fixed-size daily representation that the subsequent stage can process uniformly.

2.4.2. Stage 2: Heterogeneous Mixture Model (HetMM)

The HetMM is a probabilistic clustering model that projects each daily (variables $\times$ 48) matrix into a latent K -dimensional profile distribution. It learns a small set of *daily behavioral profiles*. A simplified conceptual explanation is that the model ‘labels’ each day with archetypes such as an *Active Day* (high morning activity, regular app usage, moderate mobility), an *Isolated Day* (low activity across all variables, restricted to a single location), an *Out-and-About Day* (high mobility, sparse app usage), and so on.

“Heterogeneous” refers to the model’s ability to mix variables with different statistical types like Gaussian-distributed step counts or Bernoulli-distributed app-usage flags. The number of profiles was set to $K = 4$, as determined in previous stages of this project by other researchers.

The output for each patient-day is a soft assignment vector $\pi_d \in [0, 1]^K$ that satisfies that $\sum_k \pi_{d,k} = 1$. This compresses high-dimensional daily behavior into a low-dimensional categorical representation. It is this sequence of daily profile assignments that the BOCPD layer monitors: a change point fires when the recent sequence of profiles deviates from the patient’s historical pattern.

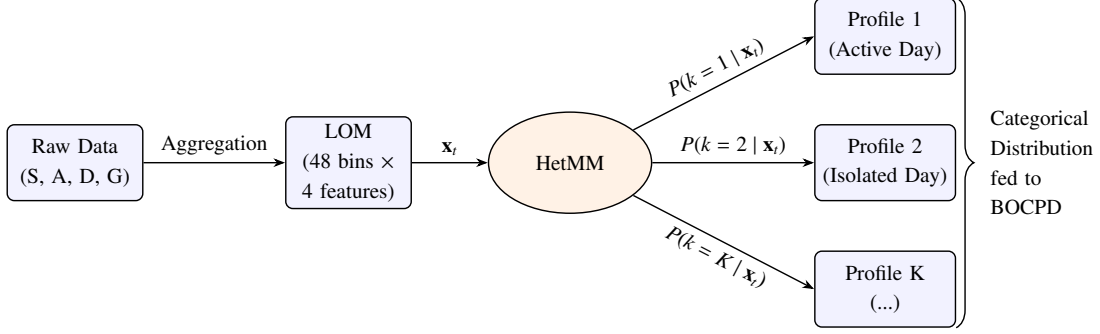


Fig. 2.1. The HetMM architecture. The Local Observation Model (LOM) creates a 48-bin matrix, which the HetMM projects into a low-dimensional distribution over K daily profiles.

2.4.3. Stage 3: UnionCPD

BOCPD gives a continuous probability that the regime has changed for each day. Rather than triggering an alarm on a single day’s probability, UnionCPD accumulates evidence over time by computing the sum of the instability probabilities over a rolling window of size W .

- **W (the detection window).** A small W (1–2 days) makes the alarm react quickly to fast changes, but at the cost of overreacting and producing more false alarms. On the other hand, a large W (5–7 days) smooths noise and captures large behavioral shifts, at the cost of losing temporal precision and missing sudden and quick behavioral shifts. This project tested $W \in \{1, 2, 5, 7\}$.
- **Threshold.** Once W is fixed, the threshold controls the trade-off between sensitivity and false alarms. Lower thresholds catch more events at the cost of more false alarms.

2.5. Bayesian Online Change-Point Detection (BOCPD)

Change-point detection (CPD) is the statistical problem of identifying the exact moments in a time series at which the underlying generative distribution changes. Given a sequence of observations $x_1, x_2, \dots, x_T$, the goal is to find the set of indices $\{\tau_1, \tau_2, \dots\}$ such that, between consecutive change points, the data is drawn from a single distribution, but after each change point the distribution is different.

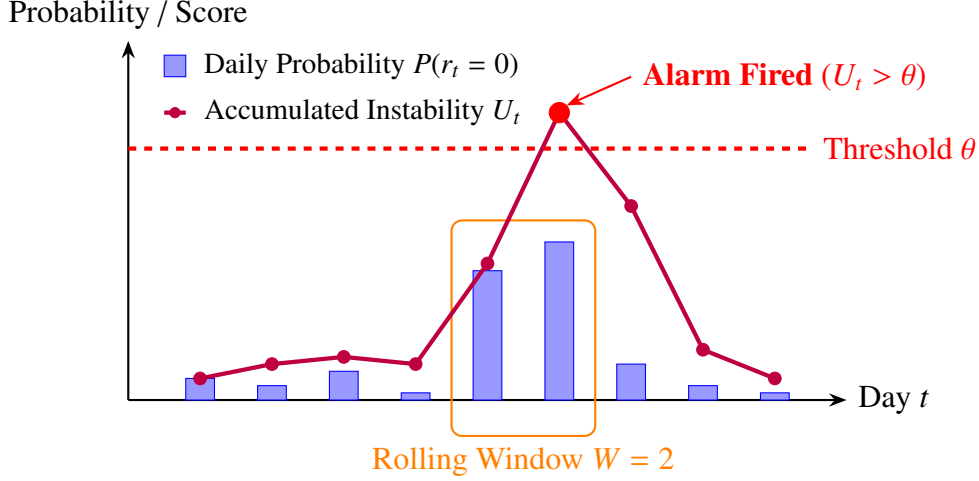


Fig. 2.2. UnionCPD operation. The system aggregates single-day BOCPD probabilities over a rolling window W (here, $W = 2$). An alarm is generated only when the accumulated instability U_t exceeds the clinical threshold θ .

In the behavioral monitoring context, the "distribution" refers to the patient's daily routine: their typical pattern of steps, app usage, mobility, etc. A change point fires when the recent days no longer look like a sample from the patient's own baseline. Therefore, this is a *per-patient* problem, where the baseline against which a deviation is measured is the patient's own historical pattern.

Two types of CPD algorithms exist:

- **Offline (retrospective) CPD** assumes the entire time series is available and searches for the optimal segmentation that minimizes some cost function.
- **Online (sequential) CPD** processes observations one at a time and emits a change-point alarm based only on data observed so far. Therefore, it is applicable to monitoring problems like the one in this thesis, where the decision to flag a patient must be made on the current day's data without knowing what tomorrow will bring.

The specific algorithm at the core of this project is Bayesian Online Change-Point Detection, originally introduced by Adams and MacKay [11]. The intuition is to maintain, at every time step, a posterior probability distribution over a single latent variable: the **run length** r_t , defined as the number of time steps since the most recent change point. If $r_t = 0$, a change point happened at time t , and if $r_t = k$, the last change point happened k steps ago.

The key is that the run-length posterior $P(r_t \mid x_{1:t})$ can be updated recursively as new observations arrive. This means that if the current run length is assumed to be r , the algorithm looks at the last r observations to form a posterior distribution over the underlying parameters (e.g., the mean and variance of the data). From this, it computes the predictive likelihood, which represents the probability of observing the new data point x_t , given that the current regime has lasted for r_{t-1} steps: $P(x_t \mid r_{t-1}, \mathbf{x}^{(r)})$.

Every time a new observation x_t arrives, the algorithm updates its belief (the posterior probability distribution) over all possible run lengths. It does this by evaluating how well each active run length's predictive model "explained" the new point.

- If x_t aligns perfectly with the recent past, the predictive likelihood $P(x_t | r_{t-1}, \mathbf{x}^{(r)})$ will be high for the existing run lengths. The consequence is that the algorithm multiplies the previous probabilities by these high likelihoods. The posterior mass shifts toward growth ($r_t = r_{t-1} + 1$). The confidence that we are deep within a stable regime increases.
- If x_t represents a sharp deviation from the recent past (e.g., a sudden drop in step count or a spike in variance), the predictive likelihood for all existing run lengths will be extremely low. The current models fail to explain the data point. As a consequence, the recursive Bayesian update shifts the remaining probability mass to the only other alternative: a change point. The probability of a new regime ($r_t = 0$) spikes.

The output of BOCPD at each time step is therefore not a binary "change" or "no change" decision but a continuous posterior over run lengths. The probability that a change point occurred at time t is captured by the mass at $r_t = 0$. This continuous output is what allows the pipeline to produce ROC curves by 'moving' a threshold over the continuous score.

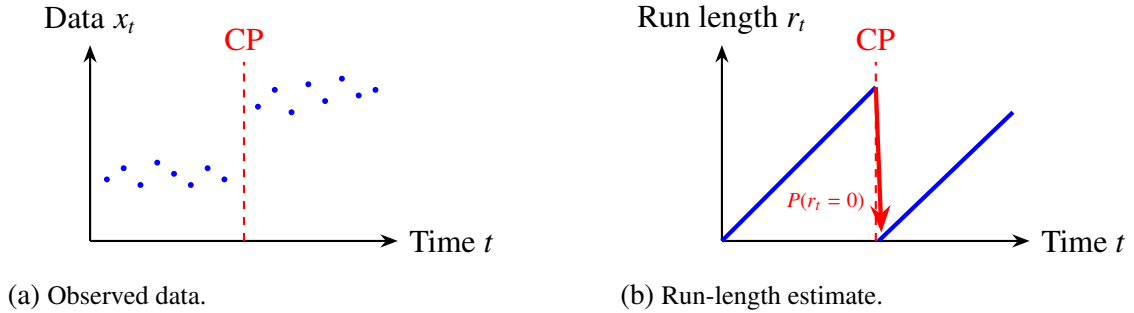


Fig. 2.3. Illustration of Bayesian Online Change Point Detection (BOCPD). A change point separates two data regimes (left). The inferred run length grows while observations remain in the same regime and resets to zero when a change point is detected (right).

Why Bayesian? There are three main reasons for adopting this framework:

1. The output is interpretable as a probability rather than an abstract test statistic.
2. The framework handles heterogeneous data types naturally; the same machinery applies whether the observations are step counts, binary app-usage flags, or categorical day-types.
3. The posterior automatically incorporates uncertainty about the patient's own baseline, which is essential when the historical record is short or contains missing data.

2.6. Metrics framework

Some of the concepts used in the evaluation metrics are non-trivial. The following sections explain the most important concepts:

2.6.1. The detection-window convention

A patient experiences an "event" on day X (the day of the suicide attempt). A change point emitted by the pipeline is counted as a **true positive** if it fires within a defined window before the event. This project originally used $[X-3, X-1]$ (a 3-day window strictly before the event day), but then other evaluation window sizes were tested: $[X-5, X-1]$ and $[X-7, X-1]$ (5-day and 7-day windows). The event day itself is excluded because an alarm triggered on day X is clinically useless, as the aggregated daily profile data is not evaluated until the end of the day.

Change points occurring outside the clinical evaluation window are counted as **false positives**, and their frequency is reported as the false positive rate (FPR) per day. To make this metric clinically interpretable, we convert the daily FPR into an expected number of alarms per month assuming a standard 30-day period: an FPR of 0.05 corresponds to 1.5 alarms/month, 0.067 equates to 2 alarms/month, 0.10 translates to 3 alarms/month, and so forth.

The FPR level is computed in the cohort of patients with no events ($TP = 0$ and $FN = 0$). The total number of valid tracking days for these patients is N . If the number of alarms raised by the model is $\sum \text{alarms}$, the FPR per day are calculated as:

$$\text{FPR}_{\text{day}} = \frac{FP}{FP + TN} = \frac{\sum \text{alarms}}{N}$$

$$\text{Alarms per month} = \text{FPR}_{\text{day}} \times 30$$

2.6.2. Sensitivity and FPR

- **Sensitivity** (true positive rate) is the fraction of true events for which at least one alarm fired within the clinical evaluation window prior to the event.
- **FPR per day** is the average rate at which false alarms fire across the no-event baseline cohort.

An important remark is that the **sample size for each metric varies**. A patient who lacks data for a variable required by a specific configuration is excluded from that configuration's FPR computation. Therefore, configurations in Takeout range from approximately 1,613 baseline-days (strict multi-variable combinations) to 3,886 baseline-days

(common single-variable combinations). For EB2, only patients that were in the Takeout cohort as well were considered, therefore resulting in a sample cohort with fewer patients: the baseline reduces to approximately 111 baseline-days across only 2 positive events.

With a Takeout baseline of 3,886 days and an EB2 baseline of 111 days, Takeout metrics can be considered good estimates, but EB2 metrics are extremely volatile. Any operating-point comparison between Takeout and EB2 should be interpreted with this asymmetry in mind.

It is important to note that the total baseline-days are aggregated across multiple, non-overlapping observation windows per patient (averaging roughly two baseline windows per no-event patient). While this maximizes the available data to robustly estimate the false positive rate, it introduces repeated measures from the same subjects. Consequently, the baseline days are not strictly statistically independent, which should be considered when interpreting the variance of the FPR estimates.

2.6.3. Signal-to-Noise Ratio (SNR)

SNR is defined as the ratio of the change-point rate inside the event window to the change-point rate outside it:

$$\text{SNR} = \frac{\text{CP rate inside clinical window}}{\text{CP rate in baseline windows}}$$

An $\text{SNR} > 1$ indicates the configuration produces alarms more frequently around events than at random. An $\text{SNR} < 1$ indicates the configuration is essentially noise. This is useful as a basic quick filter for non-informative configurations.

2.6.4. Lift over random baseline

A naive random alerter that fires uniformly with probability p per day catches an event in a k -day window with probability $1 - (1 - p)^k$. **Lift** is the ratio of actual sensitivity to this random baseline, holding p equal to the configuration's observed FPR:

$$\text{Lift} = \frac{\text{Sensitivity}}{1 - (1 - \text{FPR})^k}$$

A $\text{Lift} > 1$ indicates the configuration has genuine predictive structure. In Section 5, configurations achieving 3–5× lift are reported, which is well above what could be explained by chance.

2.6.5. Variable ablation

To quantify the contribution of an individual variable V to a configuration's performance:

$$\text{Sens_Diff}(V) = \overline{\text{Sens}}(\text{configs containing } V) - \overline{\text{Sens}}(\text{configs not containing } V)$$

Positive values mean V tends to *improve* sensitivity, whereas negative values mean V tends to *suppress* sensitivity. The analogous FPR_Diff is computed identically.

2.6.6. Variable synergy

For a pair of variables V_1, V_2 , synergy is defined as the deviation from additive expectation:

$$\text{Synergy}(V_1, V_2) = \text{Sens}(\text{both}) - [\text{Sens}(\text{only } V_1) + \text{Sens}(\text{only } V_2) - \text{Sens}(\text{neither})]$$

Positive synergy means the two variables work better together than the sum of their individual contributions would predict (i.e., they provide complementary information). Negative synergy means the variables are redundant.

2.6.7. AUC under the Hybrid FROC construction

Conventional ROC curves are not directly applicable to this problem because the "positive" events are sparse and per-patient, while false alarms are dense and per-day. We use a **hybrid FROC (Free-response ROC)** construction in which:

- **Positive instances** are per-patient-event: each event contributes a single score, and the score assigned is the maximum rolling sum of CPD instability probabilities within the clinical window $[X - ew, X - 1]$, where ew is the evaluation window length.
- **Negative instances** are per-day: each baseline day in the no-event cohort contributes one score.

Sweeping a threshold across these scores yields a curve of sensitivity (per-patient-event) versus FPR (per-day), and the area under this curve represents the hybrid FROC AUC (see Section 5).

Because negative instances are evaluated at the daily level while positive instances are evaluated at the per-event level, there is a massive asymmetry in the underlying sample sizes. Consequently, the FPR estimations in this study are statistically highly robust and stable, whereas the Sensitivity estimations are inherently more volatile and sensitive to individual patient outcomes.

2.6.8. Wilson confidence intervals

The majority of the metrics in this report use Wilson score intervals to estimate confidence bounds around proportions (like sensitivity). This choice is motivated by the fact that the Wilson construction is well-behaved at small sample sizes (as is the case in this study).

If $\hat{p}$ is the observed proportion, n is the sample size, and z is the corresponding critical value for the desired confidence level (e.g., $z = 1.96$ for 95%), the Wilson score interval is calculated as:

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}$$

Unlike the standard Wald interval ($\hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$), which is symmetric and can mathematically produce impossible bounds below 0 or above 1, the Wilson interval shifts its center toward 0.5 as the sample size n decreases. This inherently bounds the interval to the valid $[0, 1]$ range and provides a much more honest estimate of uncertainty for a cohort of $n = 31$.

2.6.9. Nested Leave-One-Patient-Out Cross-Validation

The single most important methodological tool in this report is **nested LOPOCV**. Simply selecting the "best configuration on the full dataset" suffers from overfitting because the same patients used to select the configuration are also used to evaluate it. LOPOCV corrects this:

- Train (select the champion configuration) on the other $N - 1$ patients.
- Evaluate the chosen champion on patient p alone (detected: yes/no).

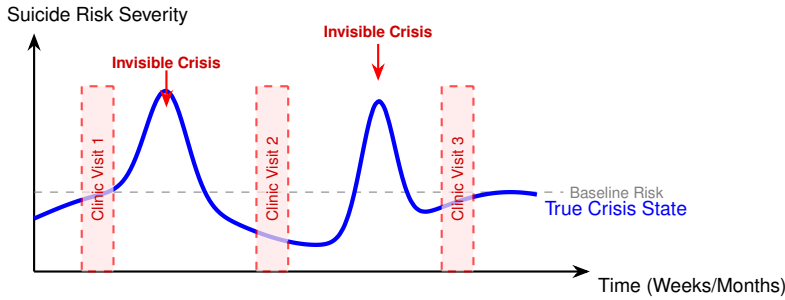
The "nested" qualifier refers to the fact that *hyperparameters* (window size W , density bracket, FPR constraint) are also re-selected on each fold's training set, not held fixed at their full-dataset optimum. Without nesting, hyperparameter overfitting alone can inflate the LOPOCV estimate substantially.

3. RELATED WORKS AND STATE OF THE ART

This section reviews the literature relevant to this project across three topics: the limitations of clinical suicide risk prediction, the application of change-point detection methods to clinical monitoring, and the use of consumer telemetry data in health research.

3.1. Limitations of clinical suicide risk prediction

Franklin et al. [1] reviewed more than 365 studies of clinical suicide prevention over fifty years and concluded that the accuracy of clinician judgment, structured risk questionnaires, and validated scales for predicting suicide attempts in the weeks-to-months window has **not improved since the 1980s**, and remains close to random for short-term window prediction (days to a week). Structured tools such as the Columbia Suicide Severity Rating Scale [12] and the Beck Scale for Suicide Ideation improve the reliability of *clinical assessment* but do not resolve the fundamental problem of *continuous monitoring*: appointments are episodic, while crisis states are transient and often invisible between appointments. This gap is the primary motivation for continuous digital monitoring systems.



3.2. Change-point detection in clinical monitoring

The original Bayesian Online Change-Point Detection algorithm was introduced by Adams and MacKay [11] and formulated a Bayesian solution to the online change-point problem. Its recursive posterior update, which is computationally cheap per time step, makes it suitable for high-frequency, long-duration monitoring sequences, such as those found when working in clinical digital phenotyping problems.

Applications of online change-point detection to health monitoring include the early detection of hospital-acquired infections from vital-sign streams [13]. In the smartphone-sensing context, Grunerbl et al. [5] used Gaussian-process change-point detection applied to GPS and phone-call features to identify depressive episodes in bipolar patients, achieving sensitivity above 70% in a small clinical cohort. This study is the closest published precursor to the present work.

3.3. Consumer telemetry data in health research

The idea of using consumer data archives for health research is recent. Coviello et al. [14] demonstrated that emotional contagion in Facebook status updates could be detected at a population scale, suggesting that consumer-grade social platforms encode mental state signals at a statistical power unavailable to small clinical studies.

In the last decade, smartphone sensing has captured the attention of researchers, and many studies have worked on the premise that the passive sensor streams already available on consumer smartphones contain clinically meaningful behavioral signals without requiring active patient effort. Saeb et al. [10] showed that GPS-derived features (location variance and circadian movement regularity collected over two weeks of daily life) correlate with PHQ-9¹ depression severity scores, with more depressed patients exhibiting lower mobility regularity and narrower location range. Canzian and Musolesi [9] demonstrated that the temporal structure of GPS mobility traces tracks transitions between depressive and non-depressive states, with irregular day-to-day mobility patterns serving as a marker of worsening mood. Wang et al. [4] extended this to longer timescales by combining passive GPS traces, phone-call logs, and screen-usage timestamps to track fine-grained depression dynamics over an academic semester, showing that behavioral signals captured without patient effort can resolve temporal mood variations that coarser weekly clinical questionnaires miss.

The common limitation of these studies is that, despite the data being collected passively, the collection platform requires patient enrollment, explicit application installation, and ongoing background permission grants. Therefore, no data is available prior to the patient engagement, making the detection algorithm unusable until several weeks later, when enough data is collected to start modeling behavior. Moreover, a patient who uninstalls the app causes data loss in exactly the same way as a patient who stops responding to mood questionnaires: the *sampling mechanism* is passive but the *participation mechanism* is not.

Google Takeout represents a different scenario. The behavioral data it contains is generated as a byproduct of ordinary smartphone use and stored by Google independently of any research activity. A patient who has never heard of digital phenotyping may already have years of this data available for export. Requesting it requires no application installation, no background permission management, and no behavioral change. This removes the participation mechanism as a structural source of missing data, making Takeout qualitatively more robust to engagement attrition than any purpose-built sensing platform. No prior study in the mental health or suicide prevention literature has evaluated whether this archive of data contains a clinically usable signal.

¹The Patient Health Questionnaire-9 (PHQ-9) is a standardized 9-item self-report scale widely used in clinical settings to screen for and measure the severity of depression.

3.4. Summary and research gap

Literature review shows that the current state of clinical research in the field is:

1. Clinical prediction of near-term suicide attempts performs near chance and has not improved in decades.
2. Active digital phenotyping can capture behavioural signals relevant to mental health but suffers severe MNAR when the signal is most needed.
3. Passive smartphone sensing can detect mood-state-level changes and, in small studies, correlates with suicidal ideation fluctuations. No study has demonstrated prospective detection of confirmed suicide attempts [15].
4. Online change-point detection has been applied to mood-episode monitoring but not to pre-attempt alarm generation.
5. Google Takeout as a clinical research data source has not been previously explored for suicide risk.

This study fills gaps 3, 4, and 5 simultaneously: it uses Google Takeout as a passive digital phenotyping data source (gap 5), applies BOCPD within the pipeline (gap 4), evaluates prospective sensitivity at clinically specified alarm budgets on a confirmed-attempt cohort (gap 3), and benchmarks the passive source against a research-grade platform (EB2) on the same cohort. The nested LOPOCV validation protocol ensures that the resulting performance estimates are prospective rather than in-sample, an aspect that is sometimes overlooked in the prior literature.

4. METHODS

4.1. Study design overview

This study uses multi-variable smartphone data from patients who experienced a confirmed suicide attempt (the *event cohort*) and patients with no recorded event during their observation window (the *no-event cohort*). The pipeline ingests patient-level data, projects each day into a latent behavioral profile via the HetMM, monitors the profile sequence via Bayesian Online Change-Point Detection, and emits per-day alarm scores.

The study compares two data sources head-to-head:

- **EB2.** A research-grade clinical sensing platform that requires active patient cooperation for some variables. EB2 contributes nine variables (S, A, D, G, V, Z, E, R, K) and represents the upper bound of what dedicated digital phenotyping can capture.
- **Google Takeout.** The consumer data archive that any Google account holder can request. Takeout contributes four behavioral variables (S, A, D, G) plus the generated K weekend flag.

4.2. Cohort and data sources

4.2.1. Observation windows

Throughout this study, the unit of analysis is the *observation window*: a contiguous 30-day period of behavioural data for a single patient. For event patients, the window ends on the day of the suicide attempt (day X), and the clinical detection window $[X-ew, X-1]$ is the subset of days within which a change-point alarm counts as a successful detection. For no-event patients, the window covers a 30-day period during which no attempt was recorded, and every day contributes to the false-alarm rate estimate. A single patient may contribute multiple windows if they experienced multiple attempts at different points in their clinical history, or multiple stable periods between clinical contacts.

4.2.2. Event cohort

The event cohort consists of **31 patients** providing a total of **52 distinct observation windows**, each with an associated confirmed suicide attempt. Because some patients experienced multiple attempts at different points in their clinical history, the cohort is defined at the window level rather than strictly at the patient level. Confirmation was established by clinical chart review and required documentation of the attempt date.

For a patient's observation window to be included, at least one of the following sufficient-data conditions had to be satisfied:

- $\geq 50\%$ data density in the last 7 days before the event, OR
- $\geq 30\%$ data density in the last 10 days before the event, OR
- $\geq 25\%$ data density in the last 20 days before the event.

These thresholds were tuned in earlier project phases. Lowering them introduces structurally missing data into the analytic cohort, but making them more strict would reduce the study cohort to an impractically small size.

4.2.3. No-event cohort

The no-event cohort consists of **37 patients** providing a total of **73 stable observation windows**. These are drawn from patients enrolled in clinical follow-up who did not record a suicide attempt during those observation windows. Because patients experience multiple stable periods throughout their treatment, almost every patient in this cohort provides two distinct observation windows. The no-event cohort provides the per-day baseline against which the false-alarm rate (alarms per month) is calibrated.

4.2.4. Within-subject cohort overlap

An important methodological characteristic of this study is its within-subject baseline design. Because cohort inclusion is determined at the window level, there is substantial overlap between the two cohorts: **30 of the 37 patients** providing baseline no-event windows are also present in the event cohort during different periods of their clinical timeline. This yields a total of **38 unique patients** across both cohorts (30 in both + 1 event-only + 7 no-event-only).

Note: one event patient is excluded from the LOPOCV analysis (Section 5.7) because no configuration in the pool had sufficient data to evaluate that patient, reducing the LOPOCV cohort to 30 patients.

4.2.5. Data density conditions

To 'decouple' the effect of data density from the actual signal present in the data, and to study how the pipeline degrades as data quality worsens, the same patient timeline is processed at four density conditions:

- **d=0.3** — artificially downsampled to retain 30% of observations per variable.

- **d=0.5** — artificially downsampled to 50%.
- **d=0.7** — artificially downsampled to 70%.
- **d=1.0 (Natural)** — the unmodified data, in which each patient keeps whatever density they naturally had.

If a patient’s natural density is already below the target, they are used as-is (no up-sampling). Each density condition produces a distinct set of experiments, and the pipeline reports per-density results.

To contextualise, Table 4.1 reports the mean natural data density of each Takeout variable across the full event cohort.

TABLE 4.1. MEAN NATURAL DATA DENSITY PER TAKEOUT VARIABLE ACROSS THE EVENT COHORT (N=31).

Variable	Mean natural density
G (Geolocation)	87.4%
A (App Usage)	71.9%
S (Steps)	67.2%
D (Distance)	47.9%

Geolocation and App Usage have the highest coverage, meaning most patients have dense records for these variables even before any artificial downsampling. Distance is notably sparser: at 47.9% mean natural density, approximately half of all 30-minute bins are missing.

4.2.6. Data acquisition and preprocessing

The end-to-end pipeline begins with the identification of eligible patients and the preprocessing of their data.

Cohort identification. Candidate event patients were identified by cross-matching patient identifiers against two clinical registries: a retrospective database of recorded suicide attempts (in SPSS format²) and a registry of visits to the hospital (emergency-care registry). Both sources were parsed to extract confirmed attempt dates, which served as the event day X for each patient.

Eligibility screening. From the candidate pool, patients were retained in the event cohort only if their Google Takeout archive satisfied at least one of the three minimum data-density conditions listed above.

Takeout data extraction. For each eligible patient, the set of CSV Google Takeout files (one per variable) was parsed into daily observation matrices. The app-usage stream

²SPSS format refers to data files compatible with the Statistical Package for the Social Sciences (SPSS) software.

records one row per application session with a UTC start timestamp, the steps stream records cumulative step counts at 5–30 minute intervals, the distance stream records the traveled distances, and the location stream records GPS coordinates with associated timestamps. Each stream is aligned to the patient’s local timezone and windowed to the 30-day period ending on the event day. Finally, a (30×48) matrix per variable per patient is produced, where rows are days and columns are 30-minute bins.

EB2 preprocessing. EB2 data was provided in per-patient CSV files with one row per day. Rather than a single column per variable, the files contain up to 48 columns per variable to represent the high-resolution 30-minute bins for behavioral streams (S, A, D, G, V). Patient-reported variables (E, emotions) were normalized within-patient. Timing variables (Z, sleep onset and duration) were normalized by converting their raw values in seconds into a fraction of a 24-hour day.

Independent Serialization and Runtime Downsampling. The final preprocessing step compresses the respective (30×48) matrices independently as pickle files (.pkl). These full-density arrays are then loaded dynamically by the main analysis pipeline, where artificial downsampling is applied on-the-fly to produce the $d \in 0.3, 0.5, 0.7, 1.0$ density conditions before being fed into the CPD model. Artificial downsampling was performed by uniformly and randomly masking entire days (setting them to missing or zero) until the exact target density was reached.

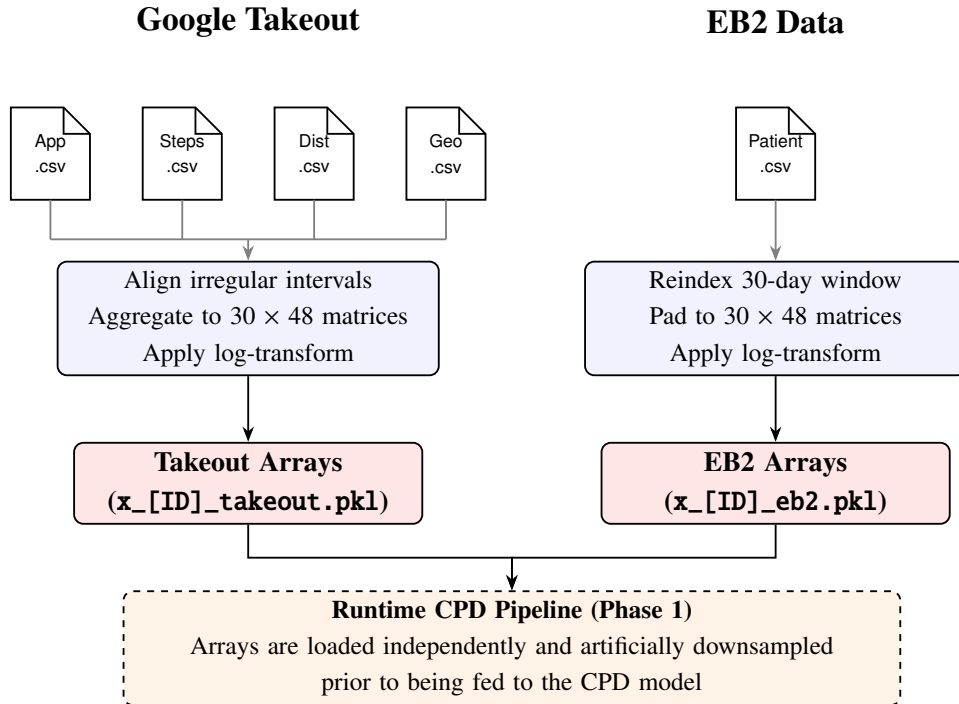


Fig. 4.1. Overview of the data preprocessing pipeline. Takeout and EB2 data are processed independently into parallel sets of serialized arrays. Downsampling is applied dynamically at runtime during the main evaluation phase.

4.3. Pipeline architecture

The three-stage pipeline is described conceptually in Section 2, and this section explains its concrete implementation.

4.3.1. Stage 1: Local Observation Model (LOM)

The LOM ingests raw event-stream data and aggregates it into 30-minute bins for each day, producing a (variables $\times$ 48 bins) matrix per patient-day.

4.3.2. Stage 2: Heterogeneous Mixture Model (HetMM)

The HetMM projects each daily (variables $\times$ 48) matrix into a latent K -dimensional profile distribution. The number of profiles K was selected in a previous stage of this project as $K = 4$. The output for each patient-day is a soft assignment vector $\pi_d \in [0, 1]^K$ with $\sum_k \pi_{d,k} = 1$.

4.3.3. Stage 3: Bayesian Online Change-Point Detection (BOCPD)

The BOCPD layer monitors the sequence of daily profile assignments. For each variable subset under evaluation, the algorithm maintains a posterior over run lengths and emits, for each day, the posterior probability that a change point occurred. These per-day probabilities are then aggregated by the UnionCPD step, where a rolling sum over a window of W days is compared to a threshold, and an alarm is emitted when the accumulated sum exceeds that threshold.

The continuous instability score (the rolling sum) is stored alongside the alarm decision, which enables the construction of the ROC curve.

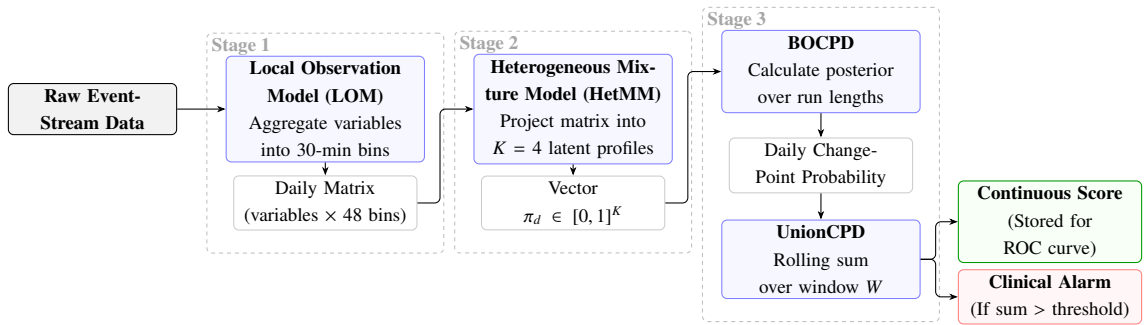


Fig. 4.2. Overview of the three-stage BehaviorCPD pipeline architecture.

4.4. Experimental design

4.4.1. Variable subset enumeration

For each data source, the pipeline tests every viable subset of available variables. A subset is considered *viable* if every variable within it passes the minimum data-density condition for that patient. For Takeout (4 candidate behavioral variables plus the K weekend flag, where K cannot be evaluated in isolation), this yields up to 30 valid subsets per patient. For EB2 (9 candidate variables), this yields up to 511 valid subsets per patient.

Subsets are tested independently per density condition and per window size. For each patient in both the event cohort (31 instances) and the no-event cohort (37 instances), the pipeline runs every subset of the variables for which that patient actually has data (processed separately for each of the two data sources). The four density conditions ($d \in \{0.3, 0.5, 0.7, 1.0\}$) are applied as independent masking passes over the same timeline, meaning each (patient, source, subset) produces four distinct evaluation rows.

This process results in 31,308 "raw" experiment records, of which 12,348 are Takeout runs and 18,960 are EB2 runs (given that EB2 has more variables to combine). These raw records are then aggregated across patients: for each unique (source, variable subset, density) combination, the sensitivity, FPR, and SNR are computed over all patients that contributed valid data for that specific configuration. Ultimately, the 31,308 patient-level raw experiments are aggregated into 4,408 distinct configurations, corresponding to 1,102 unique (source, variable subset) pairs evaluated across each of the four density levels.

4.4.2. Hyperparameter grid

Hyperparameter	Values tested	Rationale
Window size W	{1, 2, 5, 7} days	From short-term to medium-term detection horizons
Target density	{0.3, 0.5, 0.7, 1.0}	Evaluates the data-quality dependence
Evaluation window	{3, 5, 7} days	Tests how clinical-lead-time tolerance affects sensitivity
FPR constraint	{1.5, 2, 3, 4, 5, 6} alarms/month	Clinical operating-point

4.4.3. Phase structure

The pipeline executes in five sequential phases:

1. **Phase 1: Main pipeline run.** Generate a csv file containing every (patient $\times$ subset $\times$ density $\times$ window) experiment's raw CPD output. This is the basis for all the analysis.

2. **Phase 2: Missingness analysis.** Characterise the MNAR structure of each variable, trying to separate data-insufficient from model-failure false negatives.
3. **Phase 3: Hybrid FROC AUC.** Compute the AUC under the hybrid FROC construction (Section 2.6.7) for every configuration.
4. **Phase 4: Exploratory finetuning.** Grid search over $W \in \{1, 2, 5, 7\}$ and $\text{eval_window} \in \{3, 5, 7\}$ for creating a pool of champion configurations and producing per-density sensitivity at multiple operating points. These numbers are *in-sample* so they serve as an upper bound.
5. **Phase 5: Nested LOPOCV.** For each held-out patient, the champion configuration (variable subset, UnionCPD window W , and evaluation window) is re-selected on the $N - 1$ training patients and evaluated on the held-out patient. The W withdrawal threshold (0.30) is a fixed design parameter and is not re-derived per fold

4.5. Statistical analysis

Variable ablation, synergy, and density-stratified metrics are computed via the formulas in Section 2.6.5. The hybrid FROC AUC is computed by sweeping the CPD instability threshold across $[0, 1.0]$ in steps of 0.01. LOPOCV sensitivity is reported as the simple proportion of held-out patients in whom a true positive was registered.

5. RESULTS

Results are organized into sections, each addressing a specific question. Section 5.1 establishes that a behavioral signal exists in the data, Section 5.2 quantifies how strong it is, Section 5.3 shows individual variable importance, Section 5.4 documents pairwise variable interactions, Section 5.5 separates data-insufficient from model-failure false negatives, Section 5.6 presents the per-patient performance distribution, Section 5.7 provides the prospective deployment-viability assessment via LOPOCV, and Section 5.8 assesses the density coverage and clinical viability of the champion configurations.

5.1. Signal existence

Question. Is there a detectable behavioral signal in the data that is more than noise?

Finding. Yes, and it has been confirmed by different lines of evidence.

The Takeout aggregate detection rate across all configurations is **39.8%**, computed over 3,544 combinations of confirmed-event experiments (the effective sample size is 31 patients, but each appears 114 times). This is the raw number, the rate at which *some* alarm fires within the [X-3, X-1] window across every variable subset, density, and patient-event combination tested.

It is explored whether *specific* configurations exceed the random baseline at their own FPR. The lift analysis (Table 5.1) demonstrates this happens:

Configuration	Density	Sens	FPR/day	Random sens	Lift
W,K,S	0.3	57.1%	0.036	10.5%	5.4x
K,S	0.3	50.0%	0.036	10.5%	4.8x
W,S	0.3	42.9%	0.040	11.6%	3.7x
S	0.3	33.3%	0.040	11.4%	2.9x
A,K,G	0.3	75.0%	0.117	31.1%	2.4x
W,D	0.3	23.5%	0.035	10.1%	2.3x
K,S	0.7	50.0%	0.080	22.2%	2.3x

TABLE 5.1. TOP TAKEOUT CONFIGURATIONS BY LIFT OVER
RANDOM BASELINE

The top configurations achieve 3–5× lift over what a random alerter firing at the same FPR would catch.

The AUC results in Section 5.2 corroborate this: configurations achieving best AUC ranging from 0.71 to 0.90 across density brackets are inconsistent with pure-noise behavior (AUC $\approx$ 0.50).

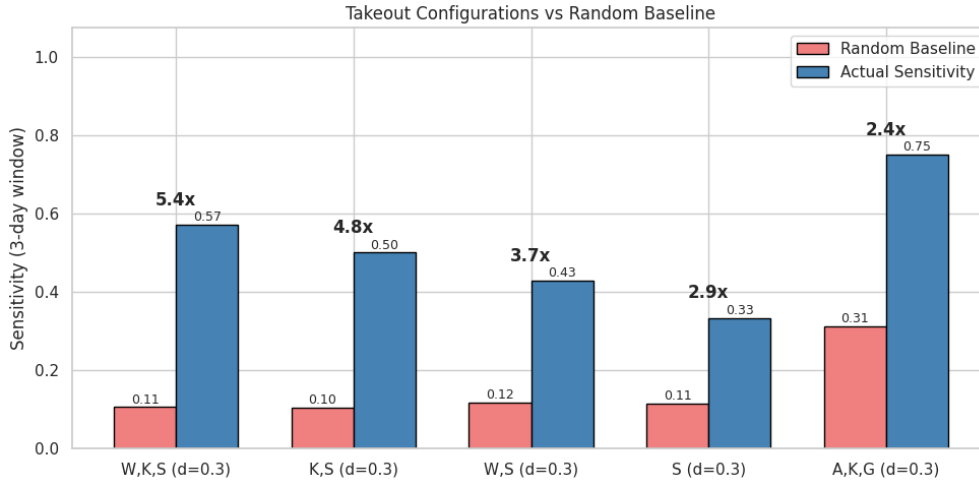


Fig. 5.1. Top Takeout configurations vs. random baseline at matched FPR (density $d=0.3$). Each configuration pair shows the random alerter sensitivity (pink) alongside the actual system sensitivity (blue) at the same FPR level.

5.2. Discrimination performance (AUC)

Question. How well does each configuration separate event windows from non-event windows?

Finding. Takeout achieves AUC up to **0.897** under the best hyperparameter combination, and $W=2$ reports as the optimal configuration *for AUC*.

5.2.1. Best AUC by data source and density

In Phase 4 finetuning, the pipeline evaluated the full grid of both EB2 and Takeout variable combinations, re-optimizing hyperparameters across all density brackets and evaluation windows.

An important caveat when analyzing these results is that the highest AUCs were achieved by EB2 configurations: combinations such as K,V,A,R,Z peaked at an AUC of 0.946 in the Natural density bracket. However, these EB2 AUC values are **not reliable estimates of genuine performance**. As noted in Section 2.6.7, the EB2 evaluation is restricted to the subset of patients common to both cohorts, which yields only ~ 2 positive events and ~ 111 baseline-days. With $n = 2$ events contributing to the sensitivity axis of the hybrid FROC curve, the AUC estimate has an enormous variance and could reach 0.946 by chance alone. Therefore, the EB2 AUC scores should not be compared directly against Takeout results.

Table 5.2 shows, highlighted in bold, that some configurations achieved excellent results using only Takeout data.

The bolded cells highlight the global maximum AUC achieved by a Takeout config-

Eval window	Density 0.3	Density 0.5	Density 0.7	Density Natural
ew=3	0.706 (K,A,D,S,G W=1)	0.802 (W,A,S,G W=2)	0.871 (K,A,S,G W=2)	0.797 (W,A,S,G W=2)
ew=5	0.771 (K,A,D,S,G W=1)	0.856 (W,A,S,G W=2)	0.897 (K,A,S,G W=2)	0.856 (K,A,D,G W=2)
ew=7	0.785 (K,A,D,S,G W=1)	0.865 (W,A,S,G W=2)	0.897 (K,A,S,G W=2)	0.868 (K,A,D,G W=2)

TABLE 5.2. BEST TAKEOUT AUC BY DENSITY AND EVALUATION WINDOW.

uration across the grid. Despite lacking the specialized clinical variables of EB2, the Takeout configurations maintain best AUC above 0.70 in every density bracket, across all conditions, peaking at 0.897 for the $d = 0.7$ bracket. This indicates that passive, consumer-grade data carries genuine and robust discriminative power, especially when integrated over short windows ($W = 2$). The following figure shows how the best results are obtained with short windows:

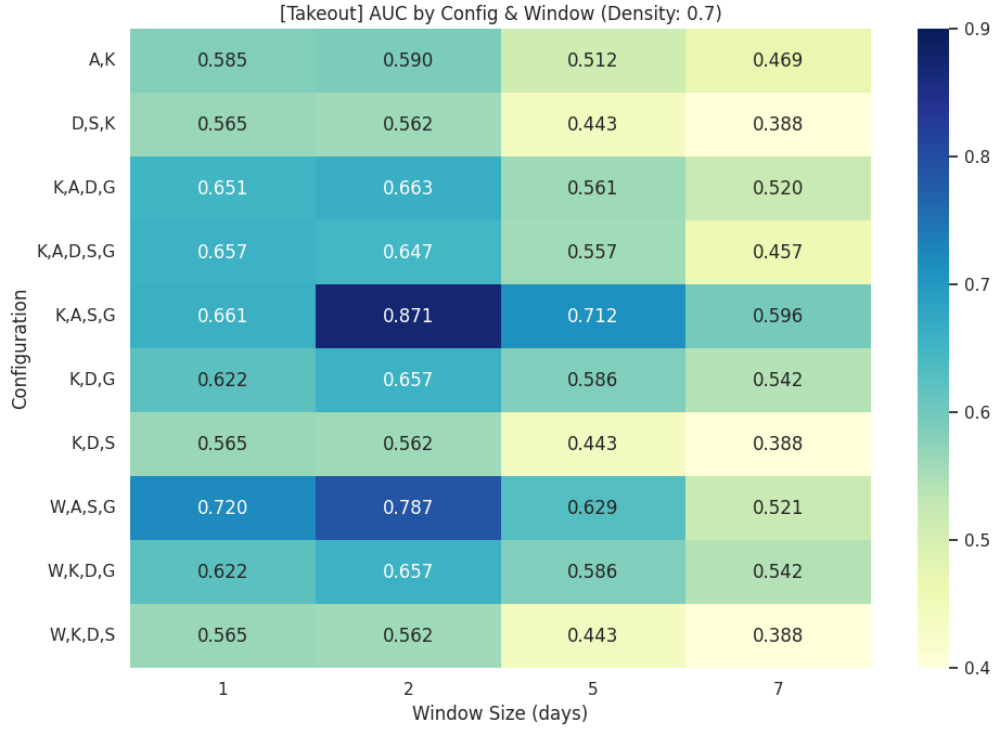


Fig. 5.2. AUC by configuration and window size at density $d=0.7$.

5.2.2. The Optimum Window size

So far, short-term windows have been shown to perform better, and the question now is what the specific optimum window size is: $W=1$ or $W=2$.

Eval window	W=1	W=2	Winner
ew=3, d=0.3	0.607	0.623	W=2
ew=3, d=0.5	0.588	0.619	W=2
ew=3, d=0.7	0.621	0.656	W=2
ew=3, d=1.0	0.614	0.691	W=2
ew=5, d=0.3	0.656	0.680	W=2
ew=5, d=0.5	0.634	0.702	W=2
ew=5, d=0.7	0.717	0.736	W=2
ew=5, d=1.0	0.684	0.747	W=2
ew=7, d=0.3	0.725	0.714	W=1
ew=7, d=0.5	0.691	0.750	W=2
ew=7, d=0.7	0.737	0.760	W=2
ew=7, d=1.0	0.746	0.771	W=2

TABLE 5.3. MEAN AUC BY WINDOW SIZE FOR TAKEOUT CONFIGURATIONS.

$W=2$ wins 11 out of 12 comparisons by mean AUC. The single exception (ew=7, d=0.3) is a 0.011 difference, which is within the margin of statistical noise.

However, when LOPOCV selects champions by sensitivity at fixed low FPR thresholds, $W=1$ "wins". The LOPOCV selector evaluates six operating points, each corresponding to a maximum tolerable false-alarm rate, where the first three are the most relevant as they are the only clinically acceptable ones (patients and doctors won't pay attention to a system that fires >5 alarms per month): 1.5 alarms/month ($\text{FPR} \leq 0.050/\text{day}$), 2/mo (≤ 0.067), 3/mo (≤ 0.100), 4/mo (≤ 0.133), 5/mo (≤ 0.167), and 6/mo (≤ 0.200).

The next table shows how often each window size was chosen across all 120 LOPOCV folds per cell (4 density brackets $\times$ 30 patients).

Alarm budget (FPR/day)	ew=3		ew=5		ew=7	
	W=1	W=2	W=1	W=2	W=1	W=2
1.5/mo (≤ 0.050)	100%	0%	100%	0%	100%	0%
2.0/mo (≤ 0.067)	100%	0%	100%	0%	100%	0%
3.0/mo (≤ 0.100)	45%	55%	98%	2%	99%	1%
4.0/mo (≤ 0.133)	48%	51%	72%	27%	53%	47%
5.0/mo (≤ 0.167)	27%	72%	49%	51%	61%	39%
6.0/mo (≤ 0.200)	27%	71%	49%	48%	28%	72%
Overall	58%	41%	78%	21%	74%	26%

TABLE 5.4. PERCENTAGE OF LOPOCV FOLDS IN WHICH EACH WINDOW SIZE WAS SELECTED AS CHAMPION AT DIFFERENT FALSE-ALARM BUDGETS.

The pattern is clear and shows that **at the clinically critical regime (≤ 2 alarms/month), $W=1$ is the optimum choice**. As the alarm budget relaxes beyond 3 alarms/month, $W=2$ becomes competitive and eventually dominant at 6 alarms/month. This is **not** a contradiction with $W=2$'s AUC superiority. $W=2$ produces a smoother ROC curve (better AUC), but $W=1$ has a higher slope at the beginning and produces sharper peaks at the strict operating points where the threshold must be set extremely high.

Therefore, the main insights are:

- **$W=2$ is optimal for overall discrimination (AUC).**
- **$W=1$ is optimal for strict operating-point deployment.**

5.2.3. Window-size effect summary

The detection horizon at a fixed evaluating window $ew=7$ shows a deterioration in mean AUC and operating-point sensitivity as W increases past 2. At the window sizes of $W=5$ and $W=7$, the system is effectively unable to operate in the clinical FPR regime:

Window size	Mean AUC	Max Sensitivity @ 1.5/mo	Max Sensitivity @ 3/mo
$W=1$	0.692	0.205	0.405
$W=2$	0.706	0.000	0.302
$W=5$	0.637	0.000	0.000
$W=7$	0.588	0.000	0.000

TABLE 5.5. COMPARISON OF AGGREGATION WINDOW SIZES.

5.3. Variable importance

Question. Which individual variables contribute most to detection sensitivity?

Finding. No individual Takeout variable alone contributes genuine discriminative signal above what its associated alarm-rate increase alone would predict (the signal is in the intersection of variables).

Variable	N with	N without	Sens with	Sens w/o	Sens_Diff	FPR_Diff
A	152	136	0.586	0.282	+0.304	+0.189
S	144	144	0.482	0.397	+0.085	+0.061
D	140	148	0.456	0.421	+0.036	+0.020
G	140	148	0.439	0.438	+0.001	+0.032
K	168	120	0.433	0.446	-0.013	-0.008

TABLE 5.6. TAKEOUT ABLATION ANALYSIS ACROSS ALL DENSITIES.

Takeout interpretation:

- **A (App Usage)** shows the largest raw Sens_Diff (+0.304), but this gain is *fully explained* by its correspondingly massive FPR increase (+0.189). A naive random alerter experiencing this same FPR increase in a 3-day clinical window would gain +0.352 in sensitivity by chance alone, which is actually higher than A's observed +0.304. Therefore, in strict isolation, A acts as a noise amplifier.
- **S and D** show modest raw Sens_Diff values that are similarly driven by their corresponding FPR increases. D is the closest to neutral (Sens_Diff +0.036, FPR_Diff +0.020).
- **G (Geolocation)** adds FPR (+0.032) without any meaningful sensitivity gain (+0.001), making it a net negative in strict isolation. However, as demonstrated later, G's true predictive value emerges exclusively through its *synergy* with D (Section 5.4).
- **K (Weekend flag)** has near-zero Sens_Diff (-0.013) and near-zero FPR_Diff (-0.008). This can be explained by the fact that K is a 'temporal control' designed to adjust the baseline expectation between weekdays and weekends, not a signal carrier meant to trigger alarms.

Methodological caveat: The ablation metric (Sens_Diff) does not hold the false positive rate constant. Consequently, it conflates genuine signal contribution with baseline alarm-rate inflation. For a strictly FPR-controlled assessment of variable importance, the Lift analysis (Section 5.1) and the hybrid FROC AUC (Section 5.2) are more reliable.

Variable	N with	N without	Sens with	Sens w/o	Sens_Diff	FPR_Diff
A	518	512	0.218	0.113	+0.105	+0.101
S	516	514	0.215	0.117	+0.098	+0.004
Z	512	518	0.188	0.145	+0.043	+0.061
V	512	518	0.186	0.146	+0.039	+0.038
R	512	518	0.182	0.150	+0.031	+0.031
G	513	517	0.167	0.165	+0.001	+0.002
K	522	508	0.152	0.180	-0.028	-0.016
D	515	515	0.133	0.199	-0.065	+0.186
E	512	518	0.111	0.220	-0.109	-0.220

TABLE 5.7. EB2 ABLATION ANALYSIS (1,030 CONFIGURATIONS).

EB2 interpretation:

- **S (Steps)** is the only variable across either data source that shows a genuine positive signal contribution after FPR correction. It increases sensitivity by 9.8 percentage points while adding virtually no false positives (+0.4pp).

- **A (App Usage)** behaves similarly to its Takeout counterpart: its high raw Sens_Diff is entirely driven by its associated FPR increase.
- **E (Emotions)** acts as a noise filter. It reduces the FPR by 22.0 percentage points. If this were just signal elimination, the sensitivity would have to drop by 39.4 percentage points. Instead, sensitivity drops only 10.9 percentage points. This means E removes approximately two thirds of the noise while preserving 72% of the signal.

5.3.1. Methodological caveat on FPR correction

It is important to note a caveat regarding the random-alterer FPR correction formula $(1 - (1 - \text{FPR})^k)$. This formula assumes daily alarms to be independent, but this does not reflect reality. CPD alarms are temporally correlated, because once the Bayesian posterior shifts, it tends to remain shifted for consecutive days. If alarms are highly correlated, the effective random gain is closer to the raw FPR_Diff itself. The true baseline should lie somewhere between these two extremes. While this does not change the main conclusions, it implies that the Takeout ablation may be more ambiguous than a strict independent-FPR correction suggests.

5.4. Variable interactions (synergy analysis)

Question. Do specific variable pairs work better together than the sum of their individual contributions would predict?

Finding. The Takeout-only synergy analysis shows that D+G is the highest-synergy pair, followed by S+G.

Pair	Both	Only_V ₁	Only_V ₂	Neither	Expected additive	Synergy
D+G	0.485	(only D) 0.429	(only G) 0.396	0.447	0.378	+0.108
S+G	0.495	(only S) 0.471	(only G) 0.392	0.403	0.460	+0.035
S+D	0.496	(only S) 0.468	(only D) 0.419	0.376	0.511	-0.014
A+G	0.568	(only A) 0.604	(only G) 0.303	0.262	0.644	-0.076
A+S	0.614	(only A) 0.563	(only S) 0.365	0.190	0.739	-0.124
A+D	0.587	(only A) 0.585	(only D) 0.346	0.201	0.730	-0.143

TABLE 5.8. TAKEOUT-ONLY PAIRWISE SYNERGY ANALYSIS
BASED ON SENSITIVITY.

Key observations and hypothetical explanations:

1. **D+G are synergistic (+0.108).** Distance and Geolocation capture complementary aspects of mobility (magnitude vs. frequency), and combining them improves detection.
2. **S+G may be mildly synergistic (+0.035).** While the effect is smaller, it is consistent with a positive interaction but cannot be statistically distinguished from zero with the current cohort size.
3. **A pairs poorly with everything.** All A-containing pairs show negative synergy. A is a high-magnitude signal that dominates configurations, leaving little marginal contribution for other variables.
4. **S+D is essentially additive (−0.014).** Both measure the same magnitude, one through stepping and the other through GPS displacement. They provide similar information and thus have no synergy.

These findings together suggest the most defensible *minimal* Takeout configuration is **D+G+K** or **S+D+G+K**, because the D+G synergy is real and the K weekend filter seems to reduce false alarms. The fact that A-containing configurations show higher raw sensitivity but with corresponding FPR penalties means that 'A' configurations are not necessarily preferable at constrained alarm budgets.

Note: This recommendation appears to conflict with the LOPOCV results (Section 5.7), where A-containing configurations (K,A,S,G and W,A,S,G) dominate champion selection. The explanation is that the synergy analysis operates on raw sensitivity without FPR constraints, while LOPOCV selects champions at fixed alarm budgets where A's noise-amplification penalty is absorbed by the threshold. At strict operating points, the LOPOCV selector values A's raw detection power despite its poor isolated synergy profile.

5.5. False-negative failure mode analysis

Question. When the pipeline misses an event, is the failure due to insufficient data or to an inability to detect the event?

Finding. Both scenarios happen.

Subset	Source	TP	FN	Sensitivity
Data-sufficient (7-day density $\geq 50\%$)	Takeout	1,021	935	52.2%
Data-sufficient (7-day density $\geq 50\%$)	EB2	893	3,667	19.6%
Data-insufficient (7-day density $< 50\%$)	Takeout	390	806	32.6%
Data-insufficient (7-day density $< 50\%$)	EB2	426	3,382	11.2%

TABLE 5.9. SENSITIVITY BY DATA SUFFICIENCY AND DATA SOURCE.

Total confirmed event experiments: 13,536 (Takeout 3,544; EB2 9,992). Of these, 6,516 are data-sufficient and 5,004 are data-insufficient. The remaining 2,016 experiments have no 7-day density record (all density columns are absent) and are excluded from this analysis.

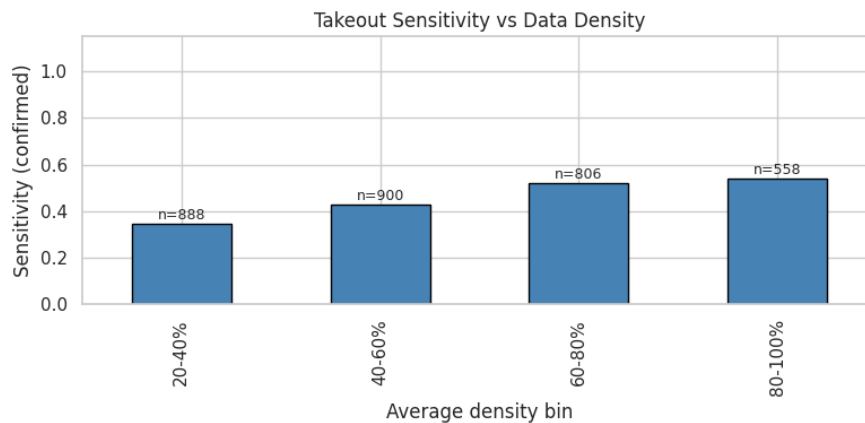


Fig. 5.3. Takeout sensitivity as a function of average 7-day data density (binned). Sensitivity increases monotonically from $\approx 33\%$ in the 20–40% density bin to $\approx 52\%$ in the 80–100% bin.

Key observations:

- Data sufficiency matters substantially within each source.** Takeout experiment-level sensitivity nearly doubles (52.2% vs. 32.6%) on the data-sufficient subset; EB2 nearly doubles too (19.6% vs. 11.2%). This effect is consistent across both sources and confirms that the pipeline’s performance degrades meaningfully when input data is sparse.
- Cross-source comparison is not possible from this table.** The 52.2% (Takeout) vs. 19.6% (EB2) figures cannot be interpreted as a comparison because:

- a) the unit of analysis is experiment-rows, not patients (each patient appears ~63 times (Takeout) or several hundred times (EB2))
- b) the false-alarm rate is not controlled. At Takeout's average FPR of 0.205/day, a random alerter in a 3-day window catches $\approx 50\%$ of events, so 52.2% represents only $\sim 1.05\times$ lift
- c) the configuration distributions differ (EB2 tests 1,030 subsets vs. Takeout's 288, including many poor-performing subsets that pull EB2's average down). A defensible source comparison would require patient-matched evaluation at a fixed operating point, which cannot be done with the current dataset.

5.6. Per-patient performance distribution

Question. Is the system uniformly mediocre across patients?

Finding. Performance depends heavily on the patient. Per-patient sensitivity has two faces, a small "detectable" subgroup that the system can detect and a larger "undetectable" subgroup that the system fails on regardless of configuration.

Patient ID	N experiments	N detected	Sensitivity	Avg density
173422	24	16	0.667	0.244
176210	24	16	0.667	0.407
179221	24	15	0.625	0.398
26920	496	294	0.593	0.575
176133	24	13	0.542	0.386
178129	72	35	0.486	0.394
45487	192	93	0.484	0.531
60690	120	57	0.475	0.569
1316	24	11	0.458	0.325
129470	424	185	0.436	0.470

TABLE 5.10. TOP-PERFORMING TAKEOUT PATIENTS RANKED BY PER-PATIENT SENSITIVITY (TOP 10 OF 31 PATIENTS).

Summary statistics:

- Patients with $\geq 50\%$ sensitivity: **5/31 (16.1%)** [95% CI: 7.1% – 32.6%]
- Patients with $< 50\%$ sensitivity: 26/31 (83.9%)
- Patients with 0% sensitivity (entirely undetectable): 1/31 (3.2%)
- Good-performing patients' avg density: 0.402
- Poor-performing patients' avg density: 0.436

Relevant finding: good-performing and poor-performing patients have similar average density (0.402 vs 0.436).

Apparent contradiction with Section 5.5: The absence of a patient-level density effect does not contradict the density–sensitivity relationship observed in Section 5.5. The difference lies in the level of analysis. At the experiment level, higher density improves the probability that a given configuration detects an event. At the patient level, however, overall data availability does not appear to determine whether any configuration can detect that patient. Patients with low density may still be detectable if their behavioural signal is strong, whereas patients with high density may remain undetectable if their pre-event behaviour shows little deviation from baseline.

5.7. Honest prospective deployment viability (LOPOCV)

Question. When champion selection is done on the N-1 patients excluding the one being evaluated, what sensitivity can the system achieve?

Finding. At ≤ 2 alarms/month the system achieves 20.0% sensitivity (6/30).

Eval	Density	1.5/mo	2/mo	3/mo	4/mo	5/mo	6/mo
ew=3	0.3	10.0% [3.5–25.6]	10.0% [3.5–25.6]	10.0% [3.5–25.6]	23.3% [11.8–40.9]	20.0% [9.5–37.3]	23.3% [11.8–40.9]
ew=3	0.5	3.3% [0.6–16.7]	13.3% [5.3–29.7]	3.3% [0.6–16.7]	13.3% [5.3–29.7]	26.7% [14.2–44.4]	33.3% [19.2–51.2]
ew=3	0.7	16.7% [7.3–33.6]	6.7% [1.8–21.3]	20.0% [9.5–37.3]	30.0% [16.7–47.9]	30.0% [16.7–47.9]	30.0% [16.7–47.9]
ew=3	Natural	3.3% [0.6–16.7]	6.7% [1.8–21.3]	3.3% [0.6–16.7]	20.0% [9.5–37.3]	20.0% [9.5–37.3]	26.7% [14.2–44.4]
ew=5	0.3	6.7% [1.8–21.3]	13.3% [5.3–29.7]	20.0% [9.5–37.3]	23.3% [11.8–40.9]	23.3% [11.8–40.9]	23.3% [11.8–40.9]
ew=5	0.5	3.3% [0.6–16.7]	3.3% [0.6–16.7]	3.3% [0.6–16.7]	13.3% [5.3–29.7]	23.3% [11.8–40.9]	26.7% [14.2–44.4]
ew=5	0.7	13.3% [5.3–29.7]	16.7% [7.3–33.6]	23.3% [11.8–40.9]	36.7% [21.9–54.5]	36.7% [21.9–54.5]	36.7% [21.9–54.5]
ew=5	Natural	3.3% [0.6–16.7]	3.3% [0.6–16.7]	6.7% [1.8–21.3]	16.7% [7.3–33.6]	20.0% [9.5–37.3]	26.7% [14.2–44.4]
ew=7	0.3	16.7% [7.3–33.6]	20.0% [9.5–37.3]	23.3% [11.8–40.9]	23.3% [11.8–40.9]	30.0% [16.7–47.9]	23.3% [11.8–40.9]
ew=7	0.5	0.0% [0.0–11.4]	6.7% [1.8–21.3]	6.7% [1.8–21.3]	26.7% [14.2–44.4]	13.3% [5.3–29.7]	23.3% [11.8–40.9]
ew=7	0.7	13.3% [5.3–29.7]	13.3% [5.3–29.7]	23.3% [11.8–40.9]	33.3% [19.2–51.2]	33.3% [19.2–51.2]	33.3% [19.2–51.2]
ew=7	Natural	6.7% [1.8–21.3]	3.3% [0.6–16.7]	3.3% [0.6–16.7]	10.0% [3.5–25.6]	20.0% [9.5–37.3]	26.7% [14.2–44.4]

TABLE 5.11. NESTED LOPOCV SENSITIVITY BY EVALUATION WINDOW, DENSITY BRACKET, AND FPR LEVEL.

Confidence intervals are Wilson 95% bounds, and bolded cells indicate the best operating point in each region of interest.

5.7.1. The clinically critical regime: ≤ 2 alarms/month

This is the regime that, when consulted, clinicians confirm can be plausibly tolerated. The results show:

- **Best at 2 alarms/month: 20.0% (6/30)** [95% CI: 9.5%–37.3%] under $ew=7$, $d=0.3$. Champion: **A,K with W=1** (selected in 30/30 LOPOCV folds).
- **Best at 1.5 alarms/month: 16.7% (5/30)** [95% CI: 7.3%–33.6%] under $ew=3$, $d=0.7$. Champion: **K,D,S with W=1** (selected in 23/30 LOPOCV folds).

This is a fundamental change in the deployment story. Both numbers have confidence intervals excluding zero. The system can catch approximately one in five attempts at a clinically realistic alarm budget. That different champion configurations emerge at different operating points (A,K at $d=0.3$ vs. K,D,S at $d=0.7$) suggests that the optimal variable subset depends on the data-density regime. That $d=0.3$ outperforms $d=0.7$ at this strict FPR constraint may reflect a mechanical effect: sparser data produces fewer change-point alarms overall, allowing the system to operate at a lower raw alarm rate without threshold adjustment.

5.7.2. The 3 alarms/month operating point

3 alarms/month was also tested to see whether a little increase in false alarms could produce a huge gain in sensitivity:

- **Best at 3 alarms/month: 23.3% (7/30)** [95% CI: 11.8%–40.9%], achieved at three distinct (ew , density) cells:
 - $ew=5$, $d=0.7$ — Champion: **W,A,S,G with W=1** (29/30 folds)
 - $ew=7$, $d=0.3$ — Champion: **A,K with W=1** (30/30 folds)
 - $ew=7$, $d=0.7$ — Champion: **K,A,S,G with W=1** (30/30 folds)

The gain from 2 to 3 alarms/month is modest (+3.3pp, from 20.0% to 23.3%), suggesting that 2 alarms/month captures most of the available signal at strict operating points. The high champion stability (29–30 of 30 folds) at all three cells confirms a clear pattern.

5.7.3. The asymptote: 5–6 alarms/month

At higher false-alarm tolerances, sensitivity flattens:

- 5 alarms/month: 36.7% (11/30) at $ew=5$, $d=0.7$. Champion: **W,A,S,G with $W=1$** (29/30 folds).
- 6 alarms/month: 36.7% (11/30) at $ew=5$, $d=0.7$ — identical patients caught; same champion.

This plateau suggests that in this range, increasing the alarm budget beyond 5/month yields no additional detections. However, this 37% ceiling is a property of the *single best (ew, density) cell*, not a structural property of the patients. As shown in Section 5.6, different operating points catch different patients: when detections are unioned across all 72 LOPOCV cells ($3 \text{ eval_windows} \times 4 \text{ densities} \times 6 \text{ alarm rates}$), 23 of 30 patients (77%) are detected at least once. Only 7 of 30 (23%) are never detected by any tested configuration. The gap between the single-cell ceiling (37%) and the union ceiling (77%) suggests that a per-patient configuration selection approach could substantially improve deployment performance beyond what any single champion achieves.

5.7.4. Density bracket effects

The clear pattern across the LOPOCV table is that the **higher density is the best when combined with evaluations windows of 5 and 7**, while **$d=0.3$ is competitive for $ew=7$** . Natural density ($d=1.0$) does not mean high density, it means no artificial downsampling was applied. Individual patients' natural densities vary (some at 0.4, others at 0.8), so this condition produces a heterogeneous cohort that a single cohort-wide threshold cannot fit well.

5.7.5. Champion stability

A key question for LOPOCV is whether the same configuration is chosen consistently. Across the four density-bracket $\times$ eval-window combinations, the champion results are:

Highly stable ($\geq 27/30$ folds agree):

- $ew=7$, $d=0.3$, 1.5–5 alarms/mo: A,K $W=1$ (30/30 in most cells)
- $ew=7$, $d=0.7$, 2–6 alarms/mo: K,A,S,G $W=1$ (30/30)
- $ew=5$, $d=0.7$, 3–6 alarms/mo: W,A,S,G $W=1$ (29/30)
- $ew=3$, $d=0.7$, 2–4 alarms/mo: K,A,S,G $W=1$ (28/30)
- $ew=3$, $d=1.0$, 3–6 alarms/mo: K,A,D,S,G $W=2$ (27/30)

Moderately stable (15–25/30 folds agree):

- $ew=3$, $d=0.3$, 3 alarms/mo: K,A,D,S,G W=2 (17/30)
- $ew=5$, $d=0.3$, 3 alarms/mo: A,K W=1 (16/30)

The high stability of the $d=0.7$ champions is the main finding for deployment: **K,A,S,G** dominates at $ew=3$ and $ew=7$ (28–30/30 folds), while the closely related **W,A,S,G** dominates at $ew=5$ (27–29/30 folds). Both share three core variables (A, S, G).

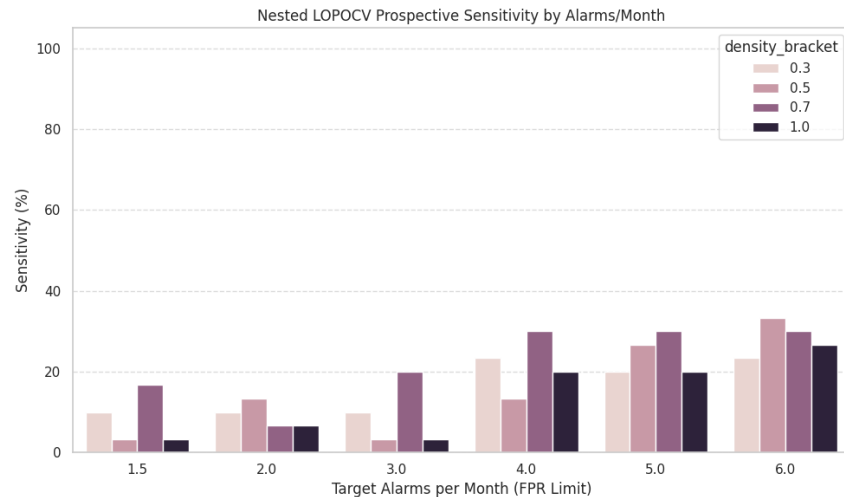


Fig. 5.4. Nested LOPOCV prospective sensitivity by target alarms per month and density bracket (eval_window=5). Each group of bars corresponds to one FPR limit (1.5 to 6 alarms/month); bars within each group are coloured by density bracket.

5.8. Density-coverage and clinical viability check

Question. Of the 38 patients in the cohort (31 event, 7 no-event), what fraction actually meet the density thresholds required for the champion configurations to operate?

Finding. Coverage exceeds 60% for both tested thresholds: 68.4% of patients meet $D \geq 0.7$, and 65.8% meet both D and $S \geq 0.3$

The exact results obtained are:

- D (Distance) overall density ≥ 0.7 : **26 out of 38 patients (68.4%)**, which is sufficient for a D-heavy champion at $d=0.7$.
- D AND S (Distance and Steps) ≥ 0.3 : **25 out of 38 patients (65.8%)**, which is sufficient for K,D,S at $d=0.3$.

Both meet the 60% coverage threshold for clinical viability. This confirms that the recommended configurations are deployable to the majority of the cohort, conditional on the cohort's demographic mix being representative of the target population.

6. DISCUSSION

This section interprets the findings in Chapter 5 in the context of the research objectives stated in Section 1.4.

6.1. Objective 1 — Prospective sensitivity and deployment envelope

The central result of the analysis round is the LOPOCV prospective sensitivity table (Section 5.7). At the clinically tolerable false-alarm level of 2 alarms per month per patient, the best Takeout-only configuration achieves **20.0% prospective sensitivity** [95% CI: 9.5% – 37.3%] under the 7-day clinical evaluation window. At 1.5 alarms per month (a slightly stricter regime), the best configuration achieves 16.7% sensitivity [95% CI: 7.3% – 33.6%] under the 3-day evaluation window.

Setting $W=1$ and extending the clinical evaluation window to 7 days produced operating points where the system catches 1/5 attempts at a tolerable false alarm level.

A 20% prospective sensitivity may not be seen as a success in isolation, but it must be evaluated against the alternatives. Compared to clinician judgment alone, which has repeatedly been shown to perform only marginally above chance for short-window prediction [1], a 20% passive sensor-based detection is a substantial improvement. Compared to active digital phenotyping platforms (which can achieve higher sensitivity but suffer from patient engagement attrition), a 20% sensitivity at the cost of zero patient effort is still a positive result.

The most defensible deployment interpretation is **complementary triage**: a 20%-sensitive passive alarm system would not replace clinical assessment, but it would identify a meaningful subset of attempts in advance of any other available signal, at no marginal cost in patient effort.

Tuning the evaluation window improved the results. Sensitivity at ≤ 2 alarms/month is zero under $ew=3$ for most density brackets, but rises to non-trivial values under $ew=5$ and $ew=7$. The system *can* detect pre-event behavioral changes, but those changes do not always cluster cleanly in the 3 days immediately before the attempt. They are sometimes spread across a week. A 7-day clinical lead time may also be clinically *more* useful as it gives more time for intervention.

Higher data density generally improves performance: Across most cells of the table, the $d = 0.7$ condition outperforms both the lower densities ($d = 0.3, d = 0.5$) and the unmodified Natural density. However, even in the lowest density brackets, a proportion of patients are still correctly detected. This suggests that while having more data is beneficial, maintaining a perfectly dense data record is neither a guarantee of success nor a strict prerequisite for detection.

The single-configuration ceiling vs. the union ceiling. At 5 and 6 alarms/month under $ew=5/d=0.7$, the system catches 11 out of 30 patients and increasing the alarm budget further does *not* add new detections. This 37% single-configuration ceiling seems to be a structural limit. However, as shown in Section 5.6, different operating points catch different patients: when detections are unioned across all 72 LOPOCV cells, 23 of 30 patients (77%) are detected at least once, and only 7 of 30 (23%) are truly never detected by any tested configuration. The gap between 37% and 77% suggests that a per-patient configuration selection approach could substantially improve deployment performance. The question of *what distinguishes the truly undetectable 23% from the rest* remains open for future work.

6.2. Objective 2 — Variable importance

The ablation analysis (Section 5.3) reveals that the strongest raw contributor, App Usage (A), shows a Sens_Diff of +0.304 but also a substantial FPR_Diff of +0.189. A random alerter gaining the same FPR increase in a 3-day window would gain +0.352 in sensitivity by chance alone, exceeding A’s actual gain. In other words, A’s apparent sensitivity advantage is fully explained by alarm-rate inflation, not genuine discriminative signal. The only variable across either source that survives this correction with a positive net contribution is Steps (S) in EB2, which adds +0.098 sensitivity with negligible FPR cost (+0.004).

This does not mean individual variables are useless, but rather that the signal emerges from specific *combinations* of variables, as demonstrated by the 3–5× lift achieved by top configurations. The synergy analysis confirms this: the D+G pair is positively synergistic (+0.108), meaning their combination captures complementary aspects of mobility (magnitude vs. frequency) that neither variable captures alone.

6.3. Objective 3 — Confidence calibration

Several findings are robust enough to be considered well-supported despite the limited cohort size:

- A signal exists in Takeout (best AUC 0.897; lift 3–5× over random).
- K is a temporal confounder control, not a signal carrier (negative Sens_Diff and negative FPR_Diff in both sources).
- At $d=0.7$, K,A,S,G ($ew=3$, $ew=7$) and W,A,S,G ($ew=5$) are stable LOPOCV champions, selected in 27–30 of 30 folds.

Nevertheless, *sample size dominates uncertainty in this study*. With $n=31$ patients, no confidence interval is narrower than about ± 20 percentage points, and many metrics are

statistically ambiguous. Any future cohort expansion will tighten these intervals substantially and may overturn findings that currently appear consistent-but-underdetermined.

6.4. Objective 4 — Is Takeout-based monitoring viable?

Takeout-based monitoring is **viable but circumscribed**. It is not a deployable clinical alarm system on its own at this maturity, but it provides genuine predictive value that no other modality can match at zero patient effort, and it is a strong candidate for a complementary triage role within a multi-modal risk-monitoring architecture.

The case for viability:

- A behavioral signal exists and is detectable above chance. AUC up to 0.897 and lift of 3–5× over a random alerter are not consistent with pure noise.
- At the clinically tolerable regime of ≤ 2 alarms/month, the system catches roughly 1 in 5 attempts (20.0%), which represents an improvement at zero patient effort.
- The champion configurations are stable across 27–30 of 30 LOPOCV folds, indicating robustness.

Caveats:

- Four out of five attempts are still missed at ≤ 2 alarms/month. The system cannot replace clinical assessment.
- At any single operating point, at most ~37% of the cohort is detected. Across all configurations, 23% of patients (7/30) are never detected regardless of settings.
- No real prospective trial has been conducted (only LOPOCV estimation), and the $n=31$ cohort produces large confidence intervals.

Recommended deployment posture:

The most defensible framing for clinical translation is as a **complementary, opt-in monitoring layer for the subset of patients who consent to Takeout export and whose clinical phenotype suggests they are in the detectable subgroup**. A patient with a history of attempts, regular smartphone use, and a relatively structured daily routine (the profile most likely to fall in the detectable subgroup) may be a strong candidate.

The system should be evaluated against the appropriate counterfactual: not “perfect prediction” but “current clinical practice augmented by an additional passive signal.” The marginal cost of running it on top of existing clinical care is almost zero, and the marginal benefit is identifying ~20% of attempts that current practice would miss.

7. CONCLUSIONS

7.1. Signal existence

A behavioural signal exists in Google Takeout data that permits pre-suicide change-point detection above chance. The best configuration (K,A,S,G, $W=2$, $d=0.7$, $ew=5$) achieves an in-sample AUC of 0.897 under the hybrid FROC construction, and top configurations achieve $2.4\times$ – $5.4\times$ lift over a random alerter at $FPR \leq 0.12$ per day.

7.2. Hyperparameter recommendations

- **For discrimination (AUC):** $W=2$ is the optimal UnionCPD window size, outperforming $W=1$, $W=5$, and $W=7$ on overall ROC-curve area.
- **For operating-point deployment:** $W=1$ is the optimal window at fixed false-alarm budgets, being selected as the champion in 58–78% of LOPOCV folds. At the clinically critical regime (≤ 2 alarms/month), $W=1$ is selected in 100% of folds.
- **Recommended variable set:** K,A,S,G at density $d \geq 0.7$ (selected in 28–30/30 folds at $ew=3$ and $ew=7$). At $ew=5$, the closely related W,A,S,G is preferred (27–29/30 folds). Both configurations share three of four variables (A, S, G).

7.3. Deployment viability

$W=1$ and a clinical evaluation window of 7 days yielded the best results at the operating level. The best nested LOPOCV prospective sensitivity estimates are:

- **≤ 2 alarms/month: 20.0% (6/30)** [95% CI: 9.5%–37.3%], at $ew=7$.
- **≤ 1.5 alarms/month: 16.7% (5/30)** [95% CI: 7.3%–33.6%], at $ew=3$.
- **≤ 3 alarms/month: 23.3% (7/30)** [95% CI: 11.8%–40.9%], at $ew=5/7$.
- **≤ 5 alarms/month: 36.7% (11/30)** [95% CI: 21.9%–54.5%], at $ew=5$.

At any single operating point, approximately 37% of the cohort is detectable and further increases in the alarm budget do not yield additional detections. However, different operating points catch different patients: across the union of all tested configurations, 77% of the cohort (23/30) is detected at least once. Only 23% (7/30) is never detected by any configuration. This suggests that per-patient configuration selection could substantially improve deployment performance.

8. LIMITATIONS AND FUTURE WORK

8.1. Sample-size limitations

The dominant source of uncertainty in this study is the cohort size of 31 confirmed-event patients. This implies:

- The “20% LOPOCV sensitivity at 2 alarms/month” finding has a CI from 9.5% to 37.3%. The true sensitivity could plausibly be anywhere in this range.
- The “37% single-configuration detection ceiling” could be as low as 22% or as high as 55% under sampling uncertainty.
- Many preliminary conclusions that are currently consistent with the data but statistically underdetermined would become clearly confirmable with a larger cohort.

Cohort expansion is the highest-priority next step for this work. A 3–5× larger event cohort (n=100–150) would tighten confidence intervals to roughly ± 8 –10 percentage points. The pipeline is ready to ingest and process new data when it arrives.

8.2. The truly undetectable patients

At any single operating point, the system detects at most $\sim 37\%$ of patients. When detections are unioned across all 72 LOPOCV cells (3 evaluation windows $\times$ 4 densities $\times$ 6 alarm rates), 23 of 30 patients (77%) are detected at least once. However, 7 of 30 patients (23%) are never detected by any tested configuration. The current study cannot determine why these patients remain undetectable. Plausible explanations include:

- **Phenotype-of-attempt differences.** Planned attempts may produce a clear pre-event behavioural signature, but impulsive attempts may not.
- **Psychiatric profile.** Patients with bipolar disorder or psychotic illness may show more dramatic behavioural shifts than those with chronic depression.
- **Medication and treatment effects.** Patients on certain medications may have suppressed behavioural signalling.

Linking clinical metadata to the analytic cohort and stratifying the detectability analysis by phenotype, diagnosis, and treatment is the second-highest priority for future work. This would answer the question of *which patients should be enrolled in Takeout-based monitoring*.

Additionally, the gap between the single-configuration ceiling (37%) and the union ceiling (77%) suggests that a substantial fraction of patients are detectable but only under specific (eval_window, density, alarm budget) combinations. Developing a per-patient configuration selection approach, in which each patient is assigned the configuration most likely to detect them, based on their baseline behavioural characteristics, could increase deployment sensitivity.

8.3. Clinical translation requirements

Before any clinical deployment, the following steps would be required:

1. **Clinical integration.** An alarm delivery mechanism (e.g., a dashboard notification to the treating clinician) must be designed, along with a response protocol defining what actions follow an alarm.
2. **Explainability.** Clinicians need to understand why an alarm fired. A post-hoc explanation layer identifying which variable(s) drove the change point should be developed.

8.4. Project roadmap

This thesis represents the first analysis cycle of an ongoing bigger project. Data collection is actively continuing and the cohort is expected to grow over time; this is the primary mechanism through which the statistical uncertainties described in Section 8.1 will be resolved.

The planned next phase encompasses two parallel tracks:

1. **Cohort expansion and model re-estimation.** As new patient data arrive, the full pipeline will be re-run and results updated. The infrastructure is already in place to ingest and process new data without modifications.
2. **Real-time alert system development.** Once the cohort reaches a sufficient size for prospective validation, the next step is to build and evaluate a real-time monitoring system that delivers alarms to treating clinicians, transitioning the pipeline from a retrospective research tool to a prospective clinical support instrument.

9. REGULATORY FRAMEWORK

This chapter discusses the main regulations and legal obligations that would apply to a clinical deployment of the system described in this thesis, in Spain and the European Union.

9.1. Personal data protection

9.1.1. General Data Protection Regulation (GDPR)

Google Takeout data contains location history, application-usage logs, and step counts. All of these variables constitute *personal data* under Article 4(1) of the General Data Protection Regulation (GDPR) [16]. Because a health-risk inference is derived from this data, it also qualifies as *data concerning health* under Article 9, which is a special category of personal data subject to heightened protection. Processing of special-category data is prohibited under Article 9(1) unless one of the exceptions in Article 9(2) applies.

Two exceptions are relevant to this work, corresponding to its two phases:

- **Research phase.** Article 9(2)(j) permits the processing of special-category data when it is necessary for scientific research purposes, provided that appropriate safeguards are implemented in accordance with Article 89(1). This is the legal basis under which the retrospective analysis reported in this thesis was conducted.
- **Clinical deployment phase.** Article 9(2)(h) permits processing that is necessary for the purposes of preventive medicine, medical diagnosis, or the provision of health treatment, provided the data is processed by, or under the responsibility of, a health professional bound by a duty of confidentiality (Article 9(3)). This would be the applicable basis for any future real-time monitoring deployment.

The key GDPR obligations that apply to this work are:

1. **Lawful basis.** A valid legal ground must be established before processing begins. For the research phase, this is Article 6(1)(e) (public interest) combined with Article 9(2)(j) (scientific research). For clinical deployment outside the context of direct medical care, explicit patient consent would be required under Article 6(1)(a) and Article 9(2)(a), meeting the conditions set out in Article 7 (freely given, specific, informed, and unambiguous).
2. **Data minimisation.** Only the variables necessary for the monitoring function (Steps, App Usage, Distance, Geolocation in the Takeout case) may be processed.

Other variables available in Takeout that do not concern the study (e.g. YouTube activity) must be excluded.

3. **Purpose limitation.** Data collected for monitoring must not be repurposed for any other use.
4. **Storage limitation and deletion.** Patient data must be retained only as long as necessary for the monitoring purpose and deleted upon patient withdrawal or end of monitoring.
5. **Data breach notification.** Any breach must be reported to the Spanish supervisory authority (AEPD) within 72 hours (Article 33) and, if high-risk, communicated to affected data subjects (Article 34).

9.1.2. Spanish national data protection law (LOPDGDD)

Spain implements the GDPR through Organic Law 3/2018, *Ley Orgánica de Protección de Datos Personales y Garantía de los Derechos Digitales* (LOPDGDD). The LOPDGDD supplements the GDPR with several Spain-specific provisions relevant to this work:

- Article 9 of the LOPDGDD establishes that, for certain special categories of data, the mere consent of the data subject is not sufficient to lift the processing prohibition; additional legal grounds are required. This is relevant because health-risk inferences from behavioural data may fall into this reinforced-protection category.
- The *Disposición Adicional Decimoséptima* (17th Additional Provision) provides specific rules for processing health data for scientific research purposes, including the permissibility of pseudonymised data when pseudonymisation is performed by someone other than the research team.
- The Data Protection Impact Assessment (DPIA) mandated by GDPR Article 35 must be conducted before any large-scale processing of health data. This is directly applicable via the GDPR's direct effect in Spain.

The AEPD (Agencia Española de Protección de Datos) is the competent supervisory authority and provides specific guidance on health data processing in digital health contexts [17].

9.2. Medical device regulation

9.2.1. EU Medical Device Regulation (MDR 2017/745)

Software that performs medical functions qualifies as a *Software as Medical Device* (SaMD) under Regulation EU 2017/745 (MDR) [18], if its output influences clinical

decisions. This is the case for the system described in this thesis, where a predicted suicide-attempt risk may trigger a safety protocol under supervision of a clinician.

Under MDR Article 51 and Annex VIII, a suicide-risk monitoring system would be classified as at least **Class IIa** (medium risk, given that it influences therapeutic decisions without being implantable) and possibly **Class IIb** if the clinical decisions influenced are high-stakes. Class IIa devices require:

1. Technical documentation conforming to Annex II, including a description of the algorithm's design, training data, validation results, and known limitations.
2. A Clinical Evaluation Report (CER) under Article 61 demonstrating clinical benefit and acceptable residual risk.
3. Conformity assessment by a Notified Body.
4. CE marking before market placement.
5. A post-market surveillance (PMS) plan and periodic Safety Update Report.

The prospective validation study required under Section 8.3 (Chapter 8) is a prerequisite for completing the CER.

9.2.2. IMDRF SaMD guidance

The International Medical Device Regulators Forum (IMDRF) has published specific guidance on SaMD classification and validation [19]. The IMDRF framework classifies SaMD using a two-dimensional matrix based on (a) the significance of the information provided (treat/diagnose, drive clinical management, or inform clinical management) and (b) the state of the healthcare situation (critical, serious, or non-serious).

For a suicide-risk monitoring system whose alarms trigger a clinical safety protocol, the most appropriate classification is “*drive clinical management*” for a “*critical*” condition (suicide being life-threatening), placing it in **Category III** of the IMDRF framework (the second-highest risk tier). This designation implies requirements for analytical validation (the algorithm's technical performance), clinical validation (prospective real-world evidence of clinical benefit), and ongoing real-world monitoring.

9.3. Clinical research ethics

9.3.1. Informed consent and IRB requirements

Data collection for the research study underlying this thesis was conducted within the ethical framework of Evidence Based Behaviour (EB2), whose data-collection protocols

have institutional ethics approval. The clinical collection and processing of data has required:

1. An application to the Institutional Review Board (IRB).
2. An informed consent form in plain language that explains: what data is collected and how, how alarms are generated and delivered, what happens when an alarm fires, who has access to the data, and the patient's right to withdraw at any time without consequences.
3. A data-sharing agreement between the research institution, the clinical institution providing patient access, and any data storage providers.

9.3.2. Vulnerability provisions

Patients being monitored for suicide risk are a *vulnerable population* under both GDPR and the Declaration of Helsinki. Specific provisions apply: the risk-benefit analysis must account for the possibility of over-reliance on algorithmic alarms, the potential for false alarms to generate distress, and the risk that a missed detection creates a false sense of security. These considerations must be documented in the Ethics Protocol and addressed in the Clinical Evaluation.

9.4. Artificial intelligence and algorithmic accountability

9.4.1. EU AI Act

The EU Artificial Intelligence Act (Regulation EU 2024/1689, which entered into force on 1 August 2024 with phased application of its provisions through August 2026) [20] classifies AI systems used in the management of health as **high-risk**. A machine-learning monitoring system such as BehaviorCPD falls in this category, and therefore must satisfy the following requirements:

1. **Transparency:** Patients must be informed that an AI system is used in their monitoring, and clinicians must be informed of the system's capabilities, limitations, and error rates.
2. **Human oversight:** No clinical decision may be made solely on the basis of the AI output; a qualified clinician must review any alarm before it triggers a clinical action.
3. **Accuracy, robustness, and cybersecurity:** The provider must demonstrate that the system performs as described under normal and foreseeable adverse conditions.

4. **Post-market monitoring:** The provider must collect and analyse real-world performance data after deployment.

The nested LOPOCV validation conducted in this study provides the prospective performance estimates that would be required for the Transparency documentation, including the acknowledged limitations that at any single operating point only ~37% of the cohort is detectable and that 23% (7/30) of patients are never detected by any tested configuration.

9.4.2. Algorithmic explainability

A complementary challenge is explainability: when an alarm fires, clinicians need to understand which behavioural signal drove the change-point detection in order to make an informed clinical decision. The current BehaviorCPD pipeline does not include an explanation layer, which constitutes a barrier to deployment under both the EU AI Act and the general principle of *meaningful human oversight*. That is why the inclusion of a post-output analysis to attribute the alarms triggered to specific variable and time contributions is a necessary next step (Chapter 8).

10. SOCIO-ECONOMIC ENVIRONMENT

This chapter analyses the budget required to execute the project, the economic and social impact of the technology, and its environmental footprint.

10.1. Project budget

The project budget covers the resources consumed during the research and development phase of this thesis. Costs are grouped into personnel, hardware, software, and indirect categories.

10.1.1. Personnel costs

Role	Hours	Rate (€/h)	Total (€)
Bachelor's student (researcher)	300	15	4,500
Thesis supervisor (senior researcher)	30	60	1,800
Clinical data curator (EB2 team)	20	40	800
Total personnel	350	—	7,100

The student-researcher estimate of 300 hours corresponds to the full project lifecycle: literature review, pipeline development, analysis, and thesis writing. The thesis supervisor estimate reflects scheduled meetings and report review. The clinical data curator cost reflects the time of the EB2 team in linking patient records and providing the anonymised data files.

10.1.2. Hardware costs

The project was executed on a personal laptop. The full pipeline runs in about 12 hours. The amortised cost of the laptop over the 9-month project period is estimated at (purchasing cost of the laptop is 530€ five years ago):

$$\text{Monthly amortization} = \frac{530}{12 \times 5} = 8.83 \text{ €/month}$$

$$\text{Project allocation: } 8.83 \times 9 = 79.47 \text{ €}$$

Item	Amortised cost (€)
Personal laptop (9/36 months)	79.47
External storage for raw data	20
Total hardware	99.47

10.1.3. Software costs

All software used in this project is open-source or free for academic use.

Tool	Licence	Cost (€)
Python 3.9 + scientific stack (NumPy, SciPy, Pandas, scikit-learn)	BSD / MIT	0
LaTeX	LPPL	0
Notion (project management)	Free tier	0
Total software	—	0

10.1.4. Indirect costs

Approximate indirect costs cover electricity and internet access.

Item	Cost (€)
Electricity + internet (9 months $\times$ €15)	135
Total indirect	135

10.1.5. Total project budget

Category	Cost (€)
Personnel	7,100
Hardware	99.47
Software	0
Indirect	135
Grand total	$\approx$ 7,300

This cost is low relative to the research it enables, and this is precisely one of the key advantages of Google Takeout-based monitoring, that no infrastructure is needed to build or maintain.

10.2. Socio-economic impact

10.2.1. The scale of the problem

Brain disorders cost Europe approximately €800 billion per year — more than cancer, cardiovascular disease, and diabetes combined — with mood disorders alone accounting for €113.4 billion annually [21]. Despite this, research investment in brain disorders remains disproportionately low relative to their economic and human burden.

Suicide is a major public health challenge in Spain and globally. According to the Spanish National Statistics Institute (INE), approximately 4,227 people died by suicide in Spain in 2022, making it the leading cause of external death, even ahead of traffic

accidents [22]. The total societal economic losses from suicide mortality in Spain have been estimated at over €2 billion per year, accounting for lost productivity and human capital costs [23]. Suicide prevention is therefore mainly a humanitarian priority, but also an economically significant one.

10.2.2. Potential impact of a deployed system

The clinical framing developed in Chapter 6 positions a Takeout-based monitor as a *complementary triage layer*. At an optimal but plausible operating point, the system could catch approximately 20% of confirmed suicide attempts with zero extra effort. In the context of Spain’s ~4,200 annual suicide deaths, even a partial deployment targeting a subpopulation of high-risk patients in clinical follow-up could translate into a meaningful number of prevented deaths.

An illustrative calculation is: if 1,000 high-risk patients in Spain were enrolled in Takeout-based monitoring, and the system catches 20% of attempts at 2 alarms/month, we can calculate the expected number of attempts detected in advance at different reattempt rates of the enrolled cohort:

At 5% reattempt rate: $1,000 \times 0.05 \times 0.20 = \mathbf{10}$ early detections/year

At 10% reattempt rate: $1,000 \times 0.10 \times 0.20 = \mathbf{20}$ early detections/year

At 15% reattempt rate: $1,000 \times 0.15 \times 0.20 = \mathbf{30}$ early detections/year

The 5–15% range reflects published annual reattempt rates for patients with a prior attempt who remain in clinical follow-up [24].

This is not a prediction of prevented deaths because detection requires intervention to prevent harm, and not all detected events would be averted, but it illustrates the scale of clinical value at modest deployment.

10.2.3. Impact on research infrastructure

Beyond the clinical value, this project demonstrates that population-scale passive monitoring research is feasible without dedicated sensing infrastructure. By validating a pipeline built entirely on consumer data exports, this work lowers the barrier to entry for research projects with tight budgets.

10.2.4. Limitations of commercial data dependency

A socio-economic risk that must be acknowledged is the current dependency on Google as a data provider. Takeout is a consumer product whose format and availability are controlled by a private company. If Google changes the export schema, discontinues the

service, or restricts data-sharing for research purposes, the pipeline would break. This commercial dependency is a long-run risk that a deployed monitoring system would need to mitigate.

10.3. Environmental impact

The environmental footprint of this project is modest. The primary energy consumption is computation: the full pipeline runs in under 12 hours on a consumer laptop (TDP $\approx 15\text{W}$), corresponding to approximately 0.18 kWh per full run. The total estimated energy consumption across all pipeline runs during the research phase (approximately 15 full runs) is $\approx 3\text{ kWh}$.

A deployment at scale would have a larger but still reasonable footprint, as it would need cloud compute for real-time monitoring, data storage and alarm delivery infrastructure.

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A. APPENDICES

A.1. Variable codebook (reference)

ID	Name	Source	Type	Behavioral meaning
S	Steps	Takeout + EB2	continuous, log-transformed	Physical activity magnitude
A	App Usage	Takeout + EB2	binary	Digital engagement
D	Distance	Takeout + EB2	continuous, log-transformed	GPS displacement magnitude
G	Geolocation	Takeout + EB2	binary	Location sensor trigger frequency
V	Activity	EB2 only	continuous	Walk/run/vehicle classifier output
Z	Sleep	EB2 only	normalized	Sleep duration and timing
E	Emotions	EB2 only	scalar	Patient-reported mood
R	Sport	EB2 only	binary	Logged exercise sessions
K	Weekend flag	Generated	binary	Weekday/weekend confounder control
W	Withdrawal flag	Generated	binary	Personal-baseline density drop indicator

A.2. Key results of the thesis

Metric	Value
Best Takeout AUC	0.897 (K,A,S,G W=2 d=0.7 ew=5)
Takeout aggregate sensitivity	39.8% [38.2 – 41.4]
Takeout data-sufficient sensitivity	52.2% (n=1,956)
EB2 data-sufficient sensitivity	19.6% (n=4,560)
LOPOCV sens @ 2 alarms/mo (best)	20.0% (6/30) [9.5 – 37.3]
LOPOCV sens @ 1.5 alarms/mo (best)	16.7% (5/30) [7.3 – 33.6]
LOPOCV sens @ 3 alarms/mo (best)	23.3% (7/30) [11.8 – 40.9]
LOPOCV sens @ 5 alarms/mo (best)	36.7% (11/30) [21.9 – 54.5]
Patients with $\geq 50\%$ per-patient sensitivity	16.1% (5/31) [7.1 – 32.6]
Detectability ceiling at any FPR	$\sim 37\%$ of cohort
D+G synergy	+0.108
MNAR gap for A	7.9 pp (62.7% vs 54.8%)
D coverage ≥ 0.7	68.4% (26/38)
K,D,S coverage ≥ 0.3	65.8% (25/38)

A.3. Reproducibility and code

The analysis pipeline is implemented in Python. Due to privacy agreements with the clinical research group, the full repository cannot be shared publicly, as it contains patient identifiers and raw clinical data protected under GDPR and the institutional data-sharing agreement. The aggregate results included in this report (CSV files and figures) can be reproduced without access to individual patient records by any collaborator with authorised access to the anonymised dataset.

Repository Main scripts

The code cannot be shared as it is proprietary work that belongs to EB2.

- `cohort_identification.py` — cross-matches patient identifiers against clinical registries and produces the event and no-event cohort CSVs.
- `identify_sufficient_takeout.py` — screens candidates against the minimum data-density thresholds.
- `run_preprocessing.py` — parses raw Takeout and EB2 CSV files, produces the (30×48) matrices per variable per patient, and serialises them to compressed pickle files.
- `main_pipeline.py` — runs every (patient $\times$ variable subset $\times$ density $\times$ window) experiment and writes `pipeline_raw_results.csv` and `pipeline_metrics.csv`.
- `extract_champions.py` — selects top-N configurations per (source, window) and writes the champion pool consumed by finetuning and LOPOCV.
- `run_finetuning.py` — grid search over W and `eval_window`, producing per-density sensitivity at multiple operating points.
- `run_lopocv.py` — executes the nested LOPOCV procedure, re-selecting the champion on $N - 1$ patients for each fold.
- `pipeline_results_analysis.py` — consumes all outputs above and produces the figures, tables, and audit reports in `results/`.

The full re-run sequence is:

```
1 python run_preprocessing.py
2 python main_pipeline.py --window 2,5,7 --target-densities
   0.3,0.5,0.7,1.0
3 python extract_champions.py
4 python run_finetuning.py --eval-windows 3,5,7
5 python run_lopocv.py --eval-windows 3,5,7
6 # Then run pipeline_results_analysis.py
```

The entire project was developed on a laptop with the following characteristics: Intel Core i3-1115G4, 8 GB RAM, Windows 11.